

FOREIGN TRADE UNIVERSITY
HO CHI MINH CITY CAMPUS

-----***-----



GRADUATION THESIS

INTERNATIONAL FINANCE

Modeling Systemic Risk and Tail Dependence in an Emerging Stock Market: Evidence from Vietnam

THESIS ID:B78

Student: Vo Tran Huong Giang

Student ID: 2211345002

Class: K61CLC2

Intake:

Supervisor: Nguyen Thi Hoang Anh, PhD

Ho Chi Minh City, tháng ... năm 20...

**FOREIGN TRADE UNIVERSITY
HO CHI MINH CITY CAMPUS**

-----***-----



GRADUATION THESIS

INTERNATIONAL FINANCE

**Modeling Systemic Risk and Tail Dependence in an
Emerging Stock Market: Evidence from Vietnam**

THESIS ID:

Student: Vo Tran Huong Giang

Student ID: 2211345002

Class: K61CLC2

Intake:

Supervisor: Nguyen Thi Hoang Anh, PhD

Ho Chi Minh City, tháng ... năm 20...

LỜI CAM ĐOAN

Tôi xin cam đoan rằng Khóa luận tốt nghiệp với đề tài “**Modeling Systemic Risk and Tail Dependence in an Emerging Stock Market: Evidence from Vietnam**” là công trình nghiên cứu do cá nhân tôi thực hiện. Toàn bộ dữ liệu, phương pháp phân tích và kết quả nghiên cứu được trình bày trong Khóa luận đều đảm bảo tính chính xác, trung thực và chưa từng được công bố trùng lặp trong bất kỳ công trình khoa học nào trước đây.

Thành phố Hồ Chí Minh, ngày tháng năm

LỜI CẢM ƠN

Khóa luận tốt nghiệp, đối với em, mang nhiều ý nghĩa. Không chỉ là học phần cuối cùng để tốt nghiệp mà còn là dấu hiệu của sự kết thúc 4 năm học đại học tại trường. Bài làm không phải là kết quả và là một chặng hành trình em đã được học tập phát triển về mặt tư duy lẫn cảm xúc, tất cả đã trưởng thành và chín mùi hơn để em hoàn thành bài. Nhưng Khóa luận này cũng không thể trọn vẹn nếu không có sự dẫn dắt, hỗ trợ, và tin tưởng từ những người xung quanh.

Em xin trân trọng bày tỏ lòng biết ơn chân thành nhất tới những người thầy người cô, gia đình, bạn bè và những cá nhân mà, nếu không có họ, ngày hôm nay em đã không ở đây và có cơ hội hoàn thành Khóa luận tốt nghiệp.

Em xin cảm ơn giảng viên hướng dẫn của em. Không chỉ dừng lại ở những góp ý quý báu cho Khóa luận tốt nghiệp, mà hơn thế, cô dạy em cách tiếp cận và làm nghiên cứu học thuật một cách bài bản và tỉ mỉ, những điều rất quan trọng cho con đường sau này.

Em xin cảm ơn các thầy cô đã từng giảng dạy những kiến thức cơ bản và chuyên ngành, những thầy cô luôn sẵn lòng lắng nghe và hỗ trợ em ở những đề tài nghiên cứu khoa học sinh viên trong quãng thời gian ở FTU2.

Con biết ơn rất nhiều ba và mẹ. Khóa luận này dành để hồi đáp tình yêu và sự ủng hộ vô điều kiện của ba mẹ dành cho con.

Mong rằng chặng đường sắp tới cũng sẽ còn những cột mốc tự hào như vậy để em được nhìn về và cảm ơn.

Sinh viên

Võ Trần Hương Giang

CONTENTS

CHAPTER 1: INTRODUCTION	3
1.1 Research Rationale.....	3
1.2 Research Background and Gap	6
1.3 Research Aim, Objectives and Questions	8
1.4 Methodology Overview	10
1.5 Scope and Limitations.....	11
CHAPTER 2: LITERATURE REVIEW	12
2.1 Systemic Risk as a Relational Phenomenon.....	12
2.2 Tail Dependence and Crisis-period Diversification Failure.....	13
2.3 Copula-based Approaches to Lower-tail Dependence.....	15
2.4 Financial Networks and Systemic Importance.....	16
2.5 Evidence from Emerging Markets, ASEAN, and Vietnam.....	18
2.6 Research Gap and Positioning of the Present Study.....	20
CHAPTER 3: METHODOLOGY	22
3.1 Research Design and Empirical Strategy.....	22
3.2 Data, Sample Construction, and Return Measurement	23
3.3 Volatility Filtering and Marginal Modelling	25
3.4 Probability Integral Transform and Pseudo-Uniform Series	26
3.5 Dependence Estimation.....	27
3.5.1 Pearson Correlation as the Benchmark.....	27
3.5.2 Clayton Copula and Lower-Tail Dependence	27
3.6 Network Construction Using Planar Maximally Filtered Graphs	29
3.7 Centrality Measures and Systemic Importance Index	29

3.8	Sector-Level Systemic Risk Measurement	31
3.9	Regime-Based Analysis	32
3.10	Framework Comparison and Crisis Validation	32
CHAPTER 4: EMPIRICAL RESULTS.....		34
4.1	From Market Composition to Tail-Risk Motivation.....	34
4.2	Regime-Dependent Lower-Tail Dependence	39
4.3	Sectoral Channels of Downside Co-movement	43
4.3.1	Intra-sector Lower-tail Clusters	43
4.3.2	Regime-specific Shifts in Intra-sector Dependence.....	45
4.3.3	Cross-sector Transmission Channels	46
4.4	Systemic Importance and Crash Transmission across Regimes.....	49
4.4.1	Tail-based Systemic Importance across Regimes.....	49
4.4.2	Systemic Hubs and Repeated Co-crash Participation.....	52
4.5	Correlation versus Lower-Tail Dependence	55
4.6	Summary of Empirical Findings.....	58
CHAPTER 5: DISCUSSION AND CONCLUSION		60
5.1	Discussion of Key Findings	60
5.2	Implications.....	62
5.3	Limitations and Future Research	66
5.4	Conclusion	67

TABLES

Table 1: Distribution of Sample Stocks by Sector	24
Table 2: Sample composition by GICS sector	34
Table 3: Descriptive statistics of daily log returns by GICS sector	36
Table 4: GJR-GARCH persistence and asymmetric volatility response by GICS sector	37
Table 5: Summary of Clayton lower-tail dependence coefficients	39
Table 6: Top intra-sector lower-tail dependent pairs	44
Table 7: Top cross-sector lower-tail dependent pairs	47
Table 8: Network Divergence and Crisis Validation	55

FIGURES

Figure 1: Overall sectoral market capitalization structure	35
Figure 2: Lower-tail dependence and market breadth across crisis regimes	40
Figure 3: Strongest intra-sector lower-tail dependent pairs	45
Figure 4: Top <i>SIItail</i> stocks across market regimes stocks	50
Figure 5: Systemic hub tickers by frequency in top 10% lower-tail dependence pairs	53

ABSTRACT

This dissertation examines systemic risk and lower-tail dependence in the Vietnamese stock market using daily data for HOSE-listed stocks from 2016 to 2025. The study applies GJR-GARCH models, Clayton copulas, and Planar Maximally Filtered Graph networks to estimate downside co-movement and identify systemically important stocks and sectors. The results show that lower-tail dependence is a significant and regime-dependent feature of the market, intensifying during major stress episodes such as the Trade War, COVID-19, the Bond Shock, and the 2025 tariff-related shock. The lower-tail network reveals an economically meaningful structure, with Financials and Real Estate forming the main transmission core through credit, collateral, margin lending, and liquidity channels. Materials, Energy, Utilities, and selected industrial stocks also become important under specific crisis regimes. Compared with Pearson correlation, the lower-tail framework captures crisis-period dependence more effectively and provides a clearer explanation of systemic risk transmission. The findings suggest that systemic risk on HOSE is best understood as a regime-dependent network of synchronized downside movements shaped by sectoral linkages and crisis-specific transmission channels.

Keywords: systemic risk, lower-tail dependence, Clayton copula, financial network, PMFG, Vietnam stock market

CHAPTER 1 INTRODUCTION

1.1 Research Rationale

Financial crises are not only periods in which stock prices decline. They are also periods in which losses become more synchronized across firms and sectors. When many stocks fall together, diversification becomes less effective precisely when investors need it most. Classical portfolio theory suggests that diversification reduces risk when asset returns are imperfectly correlated (Markowitz, 1952). However, this principle is mainly based on average co-movement. It may therefore provide an incomplete view of market risk when dependence changes during adverse market conditions. Stocks that appear only moderately connected in normal periods may become strongly linked in the lower tail, creating a structure of joint downside risk that is not fully captured by conventional correlation.

This issue is particularly relevant for emerging equity markets. Compared with developed markets, emerging markets often have thinner liquidity, higher information frictions, greater sensitivity to investor sentiment, and stronger reactions to changes in global risk appetite. These conditions can make downside co-movement more severe during periods of stress. In such markets, systemic risk is not determined only by the volatility of individual stocks, but also by the extent to which negative returns become clustered across the market. Prior studies show that tail dependence is important for measuring extreme equity risk and understanding market vulnerability during crisis periods (Duong and Huynh, 2020; Zhou and Qin, 2021). For Vietnam, where the stock market has become increasingly important for capital allocation, household investment, and corporate financing, understanding this form of downside dependence is therefore relevant for both portfolio risk management and financial-stability analysis.

Vietnam's equity market provides a meaningful setting for examining this problem. HOSE is the country's main listed equity market and includes many of the largest firms in banking, real estate, materials, consumer sectors, energy, utilities, and industrials.

These sectors are closely connected to domestic credit conditions, property-market cycles, trade exposure, commodity prices, infrastructure demand, and market liquidity. During stress periods, shocks may therefore spread across sectors through several channels. Stress in real estate stocks may affect financial stocks through credit exposure, collateral values, refinancing pressure, and investor confidence. Similarly, stress in materials or energy stocks may reflect broader concerns about construction demand, input costs, commodity prices, or external trade conditions. The key question is whether these sectoral linkages remain isolated or become channels through which downside risk is transmitted across the market.

Conventional correlation-based measures may not fully capture this type of risk. Pearson correlation measures average linear co-movement across the full return distribution, but crisis transmission is often asymmetric and state-dependent. Two stocks may not move closely together in normal periods, yet they may still experience joint extreme losses during market stress. Lower-tail dependence is designed to capture this form of relationship. From a risk-management perspective, it measures whether assets are likely to fall sharply together. From a financial-stability perspective, it indicates whether losses are becoming concentrated in a way that can weaken diversification and increase systemic fragility. This makes lower-tail dependence especially useful for studying crash transmission, where the main concern is not ordinary volatility, but joint downside movement.

Lower-tail dependence also has important implications for systemic-risk analysis. Systemic risk is not simply the risk of a large or volatile firm. It concerns the extent to which distress in one part of the market contributes to broader instability. Existing systemic-risk measures such as CoVaR, SRISK, and marginal expected shortfall emphasize the contribution of individual institutions to system-wide distress (Adrian and Brunnermeier, 2016; Acharya et al., 2017; Brownlees and Engle, 2017). Although these measures are often applied to financial institutions, the same logic can be extended to equity-market networks. Stocks or sectors that occupy central positions in a lower-tail

dependence network may act as important channels of crash-risk transmission. For Vietnam, this perspective is useful because market monitoring often focuses on price declines, volatility, and liquidity, while the structure of joint downside dependence receives less attention.

A tail-dependence network approach is therefore important for both investors and regulators in Vietnam. For investors, it helps assess whether portfolios that appear diversified under normal correlation remain protected during market stress, especially when downside co-movement becomes concentrated in sectors such as Financials, Real Estate, Materials, and Energy. This is crucial because losses during crises may arise not only from individual stock volatility, but also from the simultaneous decline of stocks linked by credit conditions, collateral values, commodity prices, market liquidity, and investor de-risking. For regulators and market supervisors, the approach helps identify which stocks and sectors form the systemic core of the market and how this core changes across shock regimes. This is particularly relevant in Vietnam, where stress can spread through domestic credit cycles, real-estate financing, corporate bond refinancing, margin lending, and external shocks such as trade disruptions, pandemic events, or sudden global repricing.

Accordingly, this study examines lower-tail dependence and sectoral systemic risk among HOSE-listed stocks. The central motivation is that market crises are not only episodes of falling prices, but also episodes in which the dependence structure of the market changes. If conventional correlation fails to identify stocks and sectors that jointly decline in the lower tail, then risk monitoring based only on average co-movement may underestimate systemic vulnerability. By focusing on lower-tail dependence and network-based systemic importance, this study aims to identify how crash risk is transmitted across Vietnam's equity market and which sectors become most important when downside risk becomes systemic.

1.2 Research Background and Gap

The study of systemic risk has increasingly shifted from the risk of individual firms to the way distress spreads across the financial system. Measures such as CoVaR, systemic expected shortfall, and SRISK were developed to capture how the distress of one institution can contribute to system-wide losses (Adrian and Brunnermeier, 2016; Acharya et al., 2017; Brownlees and Engle, 2017). This perspective is important because financial instability is rarely caused only by isolated firm-level weakness. It often emerges when losses, liquidity pressure, or investor concerns are transmitted across connected institutions and markets. However, these measures are mainly designed for financial institutions and capital shortfall analysis. For stock-market applications, especially in emerging markets, a complementary approach is needed to understand how listed firms and sectors become connected during downside market states.

Existing research on contagion and spillovers provides important evidence that market risk can spread across countries and sectors. International studies show that crisis transmission cannot be fully explained by normal-period linkages (Bekaert et al., 2014), while connectedness frameworks have been widely used to measure spillovers across financial variables (Diebold and Yilmaz, 2012). In Vietnam, previous studies have shown that the domestic stock market is exposed to external shocks, including spillovers from the U.S. and Chinese markets (Anh T. T. Tran, 2023; Chien V. Nguyen and Thuy T. Nguyen, 2022). Other Vietnam-focused studies also document systemic risk, financial connectedness, and sectoral volatility spillovers (Anh P. Nguyen et al., 2023; Nhat T. H. Dang et al., 2023; Mensi et al., 2024; Van T. T. Vu and Kham D. Tran, 2019). These findings confirm that risk transmission is relevant in Vietnam and that connectedness tends to intensify during stress periods. Nevertheless, much of this evidence remains either cross-market, institution-level, or sector-index based. It therefore provides limited evidence on which individual HOSE-listed stocks and sectors become central to domestic downside-risk transmission.

This limitation is important because crisis risk in equity markets is not only about average co-movement. During downturns, stocks may become more strongly connected in the lower tail even if their normal-period correlation is moderate. Copula methods are useful in this context because they allow dependence to be modelled separately from marginal return distributions and can capture nonlinear and asymmetric dependence (Sklar, 1959; Nelsen, 2006; Joe, 2014). Prior studies show that tail dependence is present in emerging and ASEAN equity markets (Duy Duong and Duc T. L. Huynh, 2020; Li and Rose, 2009). In Vietnam, copula-based studies have been applied to oil and gas firms and portfolio optimization (Anh T. Le and Binh T. T. Dao, 2021; Thuy T. Nguyen et al., 2020). However, these studies mainly estimate dependence or support portfolio construction. They do not fully explain how lower-tail dependence can be translated into systemic importance at the stock and sector level.

Network analysis provides the missing link between pairwise dependence and systemic interpretation. A dependence matrix can show which stock pairs are connected, but it does not directly identify which stocks act as central nodes, bridges, or channels of risk transmission. Filtered financial networks, such as the Planar Maximally Filtered Graph, help extract the most informative links from a dense market structure (Tumminello et al., 2005). Prior studies show that tail-dependence networks can differ from Pearson correlation networks and may reveal different structures of financial risk (Wen et al., 2018). Centrality-based approaches also allow systemic importance to be measured according to a stock's position in the lower-tail network (Wang and Huang, 2021). However, this integrated approach has not been widely applied to Vietnam's equity market at the stock and sector level.

The main research gap is therefore not simply that lower-tail dependence has received limited attention in Vietnam. More specifically, existing studies have not fully connected three elements: stock-level lower-tail dependence, network-based systemic importance, and sectoral crash transmission on HOSE. As a result, it remains unclear whether downside risk in Vietnam is concentrated around specific domestic transmission

channels, such as the Financials–Real Estate credit-collateral channel or the Materials–Energy industrial channel. It also remains unclear whether the stocks that appear important under average correlation are the same stocks that become important during lower-tail market stress. This distinction matters because a stock may be highly correlated with the market in normal periods without acting as a central crash transmitter, while another stock may show only moderate average correlation but become highly connected during extreme downside states.

This study addresses these gaps by constructing a lower-tail dependence network for HOSE-listed stocks and comparing it with a conventional Pearson correlation network. The study contributes to systemic-risk measurement by developing a tail-dependence-based systemic importance index for individual stocks. It contributes to dependence modelling by examining whether Clayton lower-tail dependence captures crisis co-movement differently from Pearson correlation. It also contributes to Vietnam-focused financial-stability research by identifying which sectors form persistent or regime-specific channels of downside-risk transmission. By focusing on the 2016–2025 period, the study links the empirical framework to several economically meaningful regimes, including normal periods, trade-related shocks, COVID-19, domestic funding stress, and recent tariff-related uncertainty.

1.3 Research Aim, Objectives and Questions

The aim of this study is to examine how lower-tail dependence shapes systemic downside risk and crash transmission in the Vietnamese stock market. Instead of viewing market risk only through individual stock volatility or average return co-movement, the study focuses on whether HOSE-listed stocks become jointly connected during adverse market states. The central premise is that systemic risk becomes more important when losses are concentrated across related stocks and sectors, especially when these stocks occupy central positions in the downside dependence network.

The study is built around three related issues. First, it examines whether lower-tail dependence exists among HOSE-listed stocks and whether this dependence is concentrated within specific sectors or transmitted across sectors. This helps identify where diversification benefits are most likely to weaken during market downturns. Second, it identifies which stocks and sectors become systemically important in the lower-tail network. This distinction is important because a stock may be volatile in isolation without being central to crash transmission, while another stock may become systemically important because it is strongly connected to other stocks during stress periods. Third, the study evaluates whether a lower-tail copula framework provides additional information beyond conventional Pearson correlation in capturing downside co-movement and crisis-period losses.

Based on this aim, the study has three objectives:

1. To assess the degree and sectoral structure of lower-tail dependence among HOSE-listed stocks, including both intra-sector and cross-sector downside co-movement.
2. To measure the systemic importance of individual stocks and sectors in the lower-tail dependence network, and to examine whether this importance changes across market regimes, including normal periods, the Trade War, COVID-19, the Bond Shock, and tariff-related shocks.
3. To compare tail-dependence-based systemic risk measurement with conventional Pearson correlation, in order to evaluate whether lower-tail dependence captures crisis-relevant co-movement more effectively than average linear correlation.

Accordingly, the study addresses the following research questions:

1. What is the degree and sectoral structure of lower-tail dependence among HOSE-listed stocks?

2. Which stocks and sectors contribute most to market-wide downside risk and crash transmission on HOSE, and does their systemic importance change across crisis regimes?
3. Does lower-tail copula dependence capture downside co-movement more effectively than conventional Pearson correlation?

1.4 Methodology Overview

This study uses daily stock return data for HOSE-listed firms from January 2016 to December 2025. The sample is constructed to capture the main sectoral structure of the Vietnamese equity market while ensuring sufficient liquidity and data availability. Daily logarithmic returns are used as the basis for measuring stock-level downside co-movement.

The empirical approach follows three main steps. First, the study estimates lower-tail dependence among HOSE-listed stocks to capture the extent to which stocks experience joint downside movements during adverse market conditions. This allows the analysis to focus on crash-related co-movement rather than average return correlation.

Second, the estimated lower-tail dependence relationships are translated into a financial network. This network approach helps identify which stocks and sectors are most central to downside-risk transmission. A systemic importance index is then constructed to rank stocks and sectors according to their role in the lower-tail network.

Third, the study compares the lower-tail dependence framework with a conventional Pearson correlation framework. This comparison evaluates whether lower-tail dependence provides additional information about crisis-period losses and sectoral risk transmission. The analysis is conducted for the full sample and across different market regimes, including normal periods, the Trade War, COVID-19, the Bond Shock, and tariff-related shocks. This design allows the study to examine whether systemic importance is stable over time or changes according to the nature of each shock.

1.5 Scope and Limitations

This study focuses on HOSE-listed stocks over the period 2016–2025. HOSE is selected because it is Vietnam’s main equity trading venue and includes many of the country’s most liquid and systemically relevant listed firms. The study uses daily stock return data to examine lower-tail dependence, sectoral downside co-movement, and systemic importance within the Vietnamese equity market.

The scope of the study is defined by three main boundaries. First, the analysis focuses on listed firms on HOSE rather than the entire Vietnamese equity market, including HNX and UPCoM. Second, the study uses market-based evidence from stock prices to examine downside co-movement, rather than balance-sheet exposures, ownership linkages, or credit relationships. Third, the empirical framework focuses on lower-tail dependence because the objective is to study crash transmission and systemic downside risk. These boundaries are consistent with the purpose of examining how systemic risk is reflected in equity-market dependence structures.

Sector-level interpretation is based on GICS classification, which allows the study to examine whether crash transmission is concentrated in specific sectors such as Financials, Real Estate, Materials, Energy, and other major groups on HOSE. The sample period covers both normal market conditions and several stress episodes, including the US–China Trade War, the COVID-19 crisis, the Bond Shock, and the 2025 Trump Tariff shock. This setting allows the study to assess whether lower-tail dependence and systemic importance remain stable over time or change across different shock regimes. By comparing tail-based and correlation-based dependence structures, the study evaluates whether downside-specific measures provide additional information for understanding crash transmission in Vietnam’s stock market.

CHAPTER 2 LITERATURE REVIEW

2.1 Systemic Risk as a Relational Phenomenon

Systemic risk is not only the risk that an individual firm or asset performs poorly. It refers to the possibility that distress in one part of the financial system spreads to other institutions, markets, or sectors and weakens the stability of the broader system. This distinction is important because a firm may appear manageable in isolation, but become systemically relevant when its distress is transmitted through common exposures, investor behaviour, funding conditions, or market liquidity. In this sense, systemic risk is fundamentally relational: it depends not only on the characteristics of individual firms, but also on how they are connected to the rest of the market.

The global financial crisis encouraged a shift from a microprudential view of risk toward a system-wide perspective. Earlier work shows that financial linkages can support risk sharing under normal conditions, but later studies emphasize that the same linkages may become channels of contagion when shocks are large enough. Allen and Gale (2000) argue that more complete financial networks can improve resilience by allowing liquidity risk to be shared across institutions. However, Acemoglu, Ozdaglar, and Tahbaz-Salehi (2015) show that dense financial networks may absorb small shocks but become fragile after shocks exceed a critical threshold. This suggests that interconnectedness has a dual role: it may stabilize the system in normal periods, but amplify losses during stress periods.

This argument is especially relevant for equity markets. Listed firms are connected not only through direct financial exposures, but also through common investors, sectoral fundamentals, liquidity conditions, and market sentiment. During normal periods, these connections may support price discovery and diversification. During crisis periods, however, they can transmit losses when investors rebalance portfolios at the same time, margin constraints tighten, or sentiment deteriorates across related sectors. Therefore,

stock-market contagion does not require direct contractual linkages. It can also occur through market-based channels reflected in stock returns.

Several systemic-risk measures have been developed to capture how the distress of one institution contributes to system-wide losses. Adrian and Brunnermeier (2016) propose CoVaR, Acharya et al. (2017) develop systemic expected shortfall, and Brownlees and Engle (2017) introduce SRISK. These measures are important because they move the analysis beyond standalone firm risk and toward systemic contribution. However, they are mainly designed for financial institutions and capital adequacy analysis. For stock-market applications, especially at the sectoral level, a complementary approach is needed to examine how market-based shocks spread through dependence structures among listed firms.

The connectedness literature provides a useful foundation for this perspective. Diebold and Yilmaz (2012) develop a spillover framework that measures how shocks are transmitted across financial variables and represents these relationships as a network. This approach is useful because it highlights that financial risk is transmitted through a structure of connections rather than through isolated price movements. Nevertheless, spillover and connectedness measures based on average return or volatility dynamics may not fully capture the asymmetric nature of crisis transmission. Market stress is often concentrated in adverse states, where losses become synchronized and diversification benefits weaken. This motivates the need to examine systemic risk not only as general connectedness, but also as lower-tail dependence.

2.2 Tail Dependence and Crisis-period Diversification Failure

Portfolio diversification is based on the principle that combining imperfectly correlated assets can reduce total risk. Markowitz (1952) formalizes this idea by showing that portfolio risk depends not only on the volatility of individual assets, but also on the covariance structure among them. In normal market conditions, this framework provides a foundation for portfolio allocation. However, its effectiveness may decline during

crisis periods, when assets that usually appear only moderately related begin to fall together.

This problem is central to downside-risk analysis. Crises are not average events; they are tail events. The main concern during a crisis is not ordinary co-movement, but the clustering of severe losses across firms and sectors. When lower-tail dependence is high, assets become more likely to experience extreme negative returns at the same time. In this situation, diversification may fail precisely when investors need it most. Therefore, tail dependence is important not only for portfolio management, but also for systemic-risk monitoring and financial-stability analysis.

The limitation of Pearson correlation follows from this point. Correlation summarizes average linear co-movement across the full return distribution. It does not distinguish between normal-period co-movement and crisis-period co-movement, nor does it capture the possibility that assets fall together more strongly than they rise together. A stock pair may have moderate average correlation but still show strong lower-tail dependence during market downturns. Conversely, a highly correlated pair may not necessarily be the most important channel of crash transmission. This distinction matters because systemic risk is concerned with joint losses and transmission during adverse states, rather than average co-movement in ordinary periods.

Empirical studies provide evidence that tail dependence is relevant for emerging and regional equity markets. Cuong Nguyen, Bhatti, and Henry (2017) examine tail dependence between the United States, Vietnam, and China, and find evidence of stronger left-tail dependence between the United States and Vietnam. This suggests that the Vietnamese market may be more exposed to joint crash behaviour with the U.S. market during extreme events than to joint upside movements. Duy Duong and Duc T. L. Huynh (2020) also document tail dependence among ASEAN-6 equity markets, showing that extreme co-movement is an important feature of regional market integration. These findings suggest that conventional correlation may understate downside vulnerability in emerging markets.

For the present study, lower-tail dependence is important because it connects portfolio risk with systemic risk. At the portfolio level, it indicates where diversification benefits may weaken during market downturns. At the market level, it reveals whether downside risk is concentrated within specific sectors or transmitted across sectors. If lower-tail dependence is concentrated around sectors such as Financials, Real Estate, Materials, or Energy, then crash risk is not evenly distributed across the market. Instead, some sectors may become more important channels of downside-risk transmission. This provides the economic motivation for examining lower-tail dependence among HOSE-listed stocks.

2.3 Copula-based Approaches to Lower-tail Dependence

A major challenge in studying financial dependence is that stock returns rarely follow normal and symmetric distributions. They often display skewness, fat tails, volatility clustering, and stronger reactions to negative shocks. Under these conditions, dependence cannot be fully understood through linear correlation alone. This is especially important for systemic-risk analysis, where the main concern is whether assets experience large losses together rather than whether they move together on average.

Copula models provide a useful framework for this problem because they allow the dependence structure between variables to be studied separately from their individual marginal distributions (Joe, 2014; Nelsen, 2006; Sklar, 1959). This feature is particularly useful in equity markets, where firms differ in volatility, liquidity, sectoral exposure, and investor base. By separating marginal behaviour from dependence, copulas allow researchers to examine whether assets become more connected in specific parts of the distribution, especially in the lower tail.

The literature highlights the usefulness of copulas for modelling nonlinear and asymmetric dependence in financial markets. Patton (2012) emphasizes that copula models are well suited to financial time series because they can capture forms of dependence that are not reflected in standard correlation. McNeil, Frey, and Embrechts

(2015) also stress the importance of tail dependence in quantitative risk management, particularly because extreme losses are the states in which standard assumptions may become less reliable. In this sense, copulas are relevant not only as statistical tools, but also as a way to study the economic problem of joint downside risk.

Within the copula literature, lower-tail dependence is especially relevant for crash-risk analysis. It captures the tendency of assets to experience extreme negative outcomes together. The Clayton copula is commonly used for this purpose because it focuses on lower-tail dependence. Its value lies in its ability to identify stock pairs that may become jointly vulnerable during market downturns, even when their average correlation is not particularly high. This makes it suitable for examining whether downside co-movement on HOSE is concentrated within particular stocks or sectors.

Recent studies also show that lower-tail dependence can be combined with financial networks to study systemic importance. D. Wang and Huang (2021) construct lower-tail dependence networks and use centrality-based measures to identify systemically important financial institutions. Their approach shows that lower-tail dependence can reveal both cross-sectional and time-varying differences in systemic importance. This supports the use of copula-based lower-tail dependence as a foundation for examining whether crash transmission in Vietnam is shaped by specific sectoral channels rather than by average market co-movement alone.

2.4 Financial Networks and Systemic Importance

Pairwise dependence measures are useful for identifying whether two assets move together, but they become difficult to interpret when the market contains many stocks. A stock market is not only a collection of separate bilateral relationships. It is a connected system in which some stocks may form clusters, some may act as bridges between sectors, and some may occupy central positions in the transmission of shocks. Network analysis provides a way to represent this structure by treating stocks as nodes and

dependence relationships as links. This allows the analysis to move beyond individual stock pairs and examine how risk is organized across the market as a whole.

This network perspective is important for systemic-risk measurement. A stock may be systemically relevant not because it is the most volatile or the largest by market capitalization, but because it is strongly connected to other stocks during stress periods. Similarly, a sector may become systemically important when its stocks are positioned at the centre of the market's dependence structure or when they connect otherwise separate groups of firms. In this sense, financial networks help translate dependence into economic interpretation. They allow researchers to identify hubs, clusters, and bridge nodes that may play an important role in the transmission of market stress.

Filtered network methods are commonly used because a full dependence matrix is too dense to interpret directly. Methods such as the Minimum Spanning Tree and the Planar Maximally Filtered Graph retain the most informative links while simplifying the overall structure of the market. Tumminello et al. (2005) introduce the Planar Maximally Filtered Graph as a method that preserves richer information than the Minimum Spanning Tree while keeping the network sufficiently sparse for interpretation. This is useful in financial markets because it allows researchers to identify meaningful dependence channels without being overwhelmed by all possible pairwise relationships.

The choice of dependence measure also matters for the resulting network. A network based on Pearson correlation reflects average co-movement, while a network based on lower-tail dependence reflects joint downside movement. These two structures may differ because crisis-period dependence does not necessarily follow normal-period correlation patterns. Wen, Yang, and W. Zhou (2018) show that tail-dependence networks of global stock markets can differ from Pearson correlation networks and reveal different structures of market influence and risk transmission. This supports the view that lower-tail networks may be more informative for analysing crash risk than networks based only on average correlation.

Recent studies also show that network centrality can be used to measure systemic importance. D. Wang and Huang (2021) develop centrality-based measures of systemic importance using lower-tail dependence networks. Their approach highlights that systemic importance depends not only on the strength of a stock's connections, but also on its position within the broader network. This idea is central to the present study. By combining lower-tail dependence with network centrality, the analysis can identify which HOSE-listed stocks and sectors become most relevant when downside risk is transmitted across the market.

Therefore, financial networks provide the bridge between lower-tail dependence and systemic-risk interpretation. Copula-based dependence measures identify which stocks tend to experience joint downside movements, while network analysis shows how these relationships are organized across the market. This framework is particularly useful for the Vietnamese equity market, where systemic risk may arise through sectoral channels such as the Financials–Real Estate credit-collateral linkage or the Materials–Energy industrial channel. In this study, the network approach is used to examine not only whether downside dependence exists, but also which stocks and sectors occupy central positions in crash-risk transmission.

2.5 Evidence from Emerging Markets, ASEAN, and Vietnam

Emerging markets provide a relevant setting for studying lower-tail dependence and systemic risk because their market structures often differ from those of developed markets. They tend to have lower liquidity, higher information frictions, greater retail investor participation, and stronger sensitivity to external shocks. These characteristics can make dependence structures less stable and can amplify downside co-movement during stress periods. As a result, diversification strategies based only on normal-period correlation may be less reliable when market conditions deteriorate.

Evidence from ASEAN and emerging markets supports this concern. Studies show that tail dependence is present across regional equity markets and that extreme co-

movement is an important feature of market integration and contagion (Duy Duong and Duc T. L. Huynh, 2020; Li and Rose, 2009). For Vietnam, previous research also finds that the domestic stock market is exposed to external shocks. Cuong Nguyen, Bhatti, and Henry (2017) document left-tail dependence between the US and Vietnamese stock markets, while Anh T. T. Tran (2023) shows that risk spillovers from the US market affect Vietnam's stock market dynamics. This evidence suggests that Vietnam should not be viewed as isolated from regional or global financial conditions.

Vietnam-specific studies further show that systemic risk and connectedness are important within the domestic market. Van T. T. Vu and Kham D. Tran (2019) examine systemic risk in the Vietnamese stock market, while P. A. Nguyen, N. P. Nguyen, and H. M. D. Le (2023) develop a Financial Risk Meter to assess systemic risk and network conditions in Vietnam. More recent evidence shows that Vietnamese financial institutions are highly interconnected, especially after COVID-19 and during periods of bank runs and market stress (Anh T. H. Nguyen and Thanh T. Le, 2025). Sectoral studies also show that volatility spillovers and extreme connectedness vary across industries and tend to intensify during turbulent periods (Mensi et al., 2024; Nhat T. H. Dang et al., 2023).

Taken together, this literature shows that Vietnam is exposed to both external shocks and domestic risk transmission. However, most existing studies focus on cross-market dependence, financial institutions, sector-level indices, or volatility spillovers. These approaches provide important evidence on connectedness, but they do not fully identify which individual HOSE-listed stocks and sectors become central to downside-risk transmission. This creates a need for stock-level lower-tail dependence analysis that links crash co-movement to sectoral systemic importance within the Vietnamese equity market.

2.6 Research Gap and Positioning of the Present Study

The reviewed literature shows that systemic risk is best understood as a relational phenomenon. Financial distress becomes systemic when losses are transmitted across firms, sectors, or markets through common exposures, investor behaviour, liquidity conditions, and dependence structures. The literature also shows that crises are tail events rather than average events. During market stress, diversification may weaken because assets that appear moderately related in normal periods can become strongly connected in the lower tail. This makes lower-tail dependence important for understanding crash co-movement and downside-risk transmission.

Existing studies provide important foundations for this research. Systemic-risk measures highlight the contribution of individual institutions to broader financial distress. Copula-based studies show that nonlinear and asymmetric dependence is relevant for financial markets, especially during extreme downside states. Network approaches then provide a way to translate pairwise dependence into systemic importance by identifying central nodes, clusters, and transmission channels. Together, these strands suggest that systemic downside risk should be studied not only through individual volatility or average correlation, but also through the structure of lower-tail dependence across the market.

However, several gaps remain in the context of Vietnam. First, existing Vietnam-focused studies mainly examine systemic risk in financial institutions, sector-level spillovers, or external connectedness with other markets. These studies show that risk transmission is relevant in Vietnam, but they do not fully identify which individual HOSE-listed stocks become central to domestic downside-risk transmission. Second, existing copula applications in Vietnam demonstrate the usefulness of tail dependence, but they often focus on selected sectors, cross-market relationships, or portfolio optimization. They do not fully translate lower-tail dependence into stock-level and sector-level systemic importance. Third, correlation-based or volatility-based

connectedness measures may not capture the structure of crash co-movement, especially when dependence becomes stronger in adverse market states.

The main research gap is therefore the lack of an integrated framework that connects lower-tail dependence, financial networks, and sectoral systemic importance in the Vietnamese stock market. It remains unclear whether downside risk on HOSE is concentrated around specific domestic transmission channels, such as the Financials–Real Estate credit-collateral channel or the Materials–Energy industrial channel. It also remains unclear whether the stocks that appear important under average correlation are the same stocks that become important during lower-tail market stress. This distinction matters because systemic importance may not be a fixed firm characteristic; it may depend on a stock’s position in the downside network and on the type of shock affecting the market.

The present study addresses this gap by constructing a lower-tail dependence network among HOSE-listed stocks and comparing it with a conventional Pearson correlation network. It contributes to systemic-risk measurement by identifying which stocks and sectors become central to downside market transmission. It contributes to dependence modelling by examining whether lower-tail copula dependence captures crisis co-movement differently from average linear correlation. It also contributes to Vietnam-focused financial-stability research by assessing whether systemic importance is persistent across sectors or changes across crisis regimes. In this sense, the study is positioned at the intersection of tail-dependence modelling, financial network analysis, and systemic-risk measurement in an emerging equity market.

CHAPTER 3 METHODOLOGY

3.1 Research Design and Empirical Strategy

This study adopts a market-based network approach to examine how downside risk is transmitted among HOSE-listed stocks. The empirical design follows the central argument of the dissertation: systemic risk in equity markets is not only reflected in individual stock volatility, but also in the way losses become connected across firms and sectors during stress periods. Therefore, the methodology is designed to identify whether Vietnamese stocks become jointly exposed in the lower tail, which stocks occupy central positions in this downside structure, and whether this structure differs from the one implied by conventional correlation.

The empirical strategy follows three main stages. First, stock returns are filtered to remove firm-specific volatility dynamics. This step ensures that the subsequent dependence estimates are not driven only by the fact that some stocks are individually more volatile than others. Second, lower-tail dependence is estimated to capture the probability that two stocks experience extreme negative movements together. This is the key measure of crash co-movement in the study. Third, the estimated dependence structure is transformed into a financial network, which allows the analysis to identify central stocks, sectoral clusters, and potential channels of downside-risk transmission.

The Clayton copula is used as the main dependence framework because it focuses on lower-tail co-movement. This choice is consistent with the objective of the study, which is to examine crash transmission rather than average return co-movement. Pearson correlation is retained as a benchmark. Its role is not to provide the main measure of systemic risk, but to test whether a conventional average-dependence framework identifies the same stocks and sectors as the lower-tail framework. This comparison is important because the dissertation asks whether correlation-based monitoring may overlook crisis-relevant dependence structures.

The methodology therefore connects the technical model to the economic question of the paper. GJR-GARCH filtering prepares stock returns for dependence estimation. The Clayton copula measures joint downside risk. The PMFG network extracts the most important dependence links. Network centrality measures identify which stocks are systemically important in the lower-tail structure. Finally, regime-based analysis and crisis validation assess whether the identified systemic importance is economically meaningful during periods of market stress.

3.2 Data, Sample Construction, and Return Measurement

The study uses daily closing prices of HOSE-listed stocks from January 2016 to December 2025. HOSE is selected because it is Vietnam's main equity trading venue and contains many of the largest and most liquid listed firms in the market. The focus on HOSE also improves data consistency, since firms listed on this exchange generally have better trading continuity than firms on smaller or less liquid market segments.

The sample is constructed to balance market representativeness and empirical reliability. First, stocks are required to have a sufficiently long and continuous trading history over the sample period. This condition is necessary because lower-tail dependence estimation requires comparable return histories across normal and crisis regimes. Second, stocks are selected within each GICS sector based on both market capitalization and liquidity. This ensures that the final sample captures firms that are economically important and actively traded, rather than firms that are large but illiquid or firms with insufficient trading observations.

After applying these filters, the final sample consists of 193 stocks across 11 GICS sectors. Sector classification is preserved throughout the analysis because the study is concerned not only with stock-level systemic importance, but also with sector-level channels of downside-risk transmission. This is particularly important for Vietnam, where sectors such as Financials, Real Estate, Materials, Energy, and Consumer Staples

are connected to different economic mechanisms, including credit conditions, collateral values, commodity exposure, domestic consumption, and external trade.

Daily logarithmic returns are calculated as follows:

$$r_{i,t} = \ln \left(\frac{P_{i,t}}{P_{i,t-1}} \right) \times 100,$$

where $P_{i,t}$ is the closing price of stock i on day t . Log returns are used because they are standard in financial risk modelling and allow price changes to be analyzed consistently over time. From an economic perspective, daily stock returns reflect how market participants incorporate information about firm fundamentals, sectoral conditions, liquidity, and macro-financial shocks. Using stock-level returns allows the study to examine how downside shocks are transmitted within the Vietnamese equity market.

Table 1: Distribution of Sample Stocks by Sector

Sector	N	%	Tickers
Real Estate	41	21.24	CCL, D2D, DIG, DTA, DXG, EVG, FDC, HAR, HDC, HQC, IJC, ITC, KBC, KDH, KPF, LDG, LGL, LHG, NBB, NHA, NLG, NTC, NTL, NVL, PDR, PTL, QCG, SCR, SGR, SJS, SZL, TCH, TDC, TDH, TIP, TIX, VIC, VPH, VPI, VRC, VRE
Cons. Staples	35	18.13	AAM, ABT, ACL, AFX, AGM, ANV, ASM, BBC, BHN, CLC, CMX, DAT, DBC, FMC, HAG, IDI, KDC, LAF, LIX, LSS, MCH, MSN, NAF, NSC, OGC, PAN, PIT, SAB, SBT, SMB, SSC, TCO, VCF, VHC, VNM
Financials	30	15.54	ACB, AGR, APG, BIC, BID, BMI, BSI, BVH, CTG, CTS, EIB, FTS, HCM, LPB, MBB, MIG, ORS, PGI, SHB, SSI, STB, TVB, TVS, VCB, VCI, VDS, VIB, VIX, VND, VPB
Industrials	26	13.47	BCG, BMP, CII, CTD, CTI, DPG, FCN, GEX, GMD, HAH, HHV, HVN, LCG, PC1, SAM, SCS, SHI, STG, TLG, TMS, TV2, VCG, VGC, VJC, VOS, VSC
Materials	19	9.84	AAA, BFC, CSV, CVT, DCM, DGC, DHC, DPM, DPR, HPG, HSG, HT1, KSB, NHH, NKG, PHR, PTB, SMC, TRC
Utilities	18	9.33	BWE, CHP, CNG, GAS, GEG, HDG, HNA, KHP, KOS, NT2, PGD, PPC, REE, SHP, SJD, TDM, VPD, VSH
Cons. Disc.	12	6.22	CTF, DGW, DRC, GIL, HAX, HHS, MWG, PET, PNJ, TCM, TSC, TTF
Info Tech	5	2.59	CMG, DLG, ELC, FPT, ITD
Energy	4	2.07	PLX, PVD, PVP, PVT
Health Care	2	1.04	DCL, DHG
Media Svc	1	0.52	CTR
Total	193	100.00	

3.3 Volatility Filtering and Marginal Modelling

Before estimating dependence between stocks, the study first filters each individual return series. This step is necessary because financial returns usually exhibit volatility clustering, fat tails, and asymmetric responses to negative shocks. These features are especially relevant in emerging markets, where investor sentiment, liquidity conditions, and market microstructure can produce abrupt changes in volatility.

If dependence were estimated directly from raw returns, the results could partly reflect stock-specific volatility rather than genuine cross-stock connectedness. For example, a highly volatile real estate or securities stock may appear strongly connected to other stocks simply because its own volatility is unstable. To reduce this problem, the study models each stock's marginal volatility process before estimating dependence across stocks.

For each stock i , the return innovation is written as:

$$r_{i,t} = \mu_i + \varepsilon_{i,t},$$

$$\varepsilon_{i,t} = \sigma_{i,t} z_{i,t},$$

where μ_i is the conditional mean, $\varepsilon_{i,t}$ is the return innovation, $\sigma_{i,t}$ is conditional volatility, and $z_{i,t}$ is the standardized innovation. The conditional variance is estimated using a GJR-GARCH(1,1) model:

$$\sigma_{i,t}^2 = \omega_i + \alpha_i \varepsilon_{i,t-1}^2 + \gamma_i \varepsilon_{i,t-1}^2 \mathbf{1}_{\{\varepsilon_{i,t-1} < 0\}} + \beta_i \sigma_{i,t-1}^2.$$

The GJR-GARCH model is appropriate because it allows volatility to respond differently to negative and positive shocks. This is important for equity markets because bad news often increases uncertainty, margin pressure, and liquidity demand more strongly than good news of similar size. In the context of systemic risk, this asymmetry matters because downside shocks are more relevant for crash transmission than ordinary return fluctuations.

The innovations are assumed to follow a skewed- t distribution to account for fat tails and asymmetry in stock returns. After the model is estimated, standardized residuals are obtained as:

$$\hat{z}_{i,t} = \frac{\hat{\varepsilon}_{i,t}}{\hat{\sigma}_{i,t}}.$$

These standardized residuals represent return innovations after controlling for time-varying volatility. Economically, this step separates individual volatility risk from systemic dependence risk. The study is not primarily interested in whether a stock is volatile on its own, but whether it remains connected to other stocks after its own volatility dynamics have been filtered out.

3.4 Probability Integral Transform and Pseudo-Uniform Series

After volatility filtering, the standardized residuals are transformed into pseudo-uniform variables using a rank-based probability integral transform:

$$u_{i,t} = \frac{\text{rank}(\hat{z}_{i,t})}{T + 1},$$

where T is the number of observations. This transformation maps each return innovation series into the interval $(0,1)$.

The purpose of this step is to make stocks comparable before estimating dependence. HOSE-listed firms differ substantially in volatility, liquidity, sector exposure, and distributional characteristics. The probability integral transform places these different return series on a common probabilistic scale. This allows the copula model to focus on the dependence structure between stocks rather than on differences in their individual return distributions.

In economic terms, this step helps answer a cleaner question: after controlling for each stock's own volatility behaviour, do two stocks still tend to experience downside shocks together? This is the relevant question for lower-tail systemic risk.

3.5 Dependence Estimation

3.5.1 Pearson Correlation as the Benchmark

Pearson correlation is used as the benchmark dependence measure. It captures average linear co-movement between two standardized return innovation series. For each stock pair (i, j) , the correlation coefficient is calculated as:

$$\rho_{ij} = \frac{\sum_t (\hat{z}_{i,t} - \bar{z}_i)(\hat{z}_{j,t} - \bar{z}_j)}{\sqrt{\sum_t (\hat{z}_{i,t} - \bar{z}_i)^2} \sqrt{\sum_t (\hat{z}_{j,t} - \bar{z}_j)^2}}.$$

Pearson correlation is included because it is widely used in portfolio allocation, risk management, and financial network analysis. It represents the conventional view of diversification: assets with lower average correlation are expected to provide better risk reduction.

However, correlation has an important limitation for this study. It treats upside and downside co-movement symmetrically and summarizes dependence across the whole return distribution. As a result, it may miss cases in which stocks are only moderately related in normal periods but become strongly connected during extreme downside states. For this reason, correlation is used as a comparison framework rather than the main measure of systemic downside risk.

For network construction, Pearson correlation is converted into a distance measure:

$$d_{ij} = \sqrt{2(1 - \rho_{ij})}.$$

This transformation allows strongly correlated stocks to be represented as closer nodes in the correlation-based financial network.

3.5.2 Clayton Copula and Lower-Tail Dependence

The main dependence measure of the study is lower-tail dependence estimated using the Clayton copula. The bivariate Clayton copula is defined as:

$$C(u_i, u_j; \theta) = (u_i^{-\theta} + u_j^{-\theta} - 1)^{-1/\theta}, \quad \theta > 0.$$

The Clayton copula is selected because it captures dependence in the lower tail. This makes it suitable for the research question of the dissertation, which focuses on whether stocks experience extreme losses together. In a systemic-risk context, high lower-tail dependence means that distress in one stock is more likely to coincide with distress in another stock. This is the condition under which diversification becomes less effective and market risk becomes more systemic.

The Clayton dependence parameter is estimated using Kendall's tau:

$$\hat{\theta}_{ij} = \frac{2\hat{\tau}_{ij}}{1 - \hat{\tau}_{ij}},$$

where $\hat{\tau}_{ij}$ is Kendall's tau between $u_{i,t}$ and $u_{j,t}$. The lower-tail dependence coefficient is then obtained as:

$$\lambda_{ij}^L = 2^{-1/\hat{\theta}_{ij}}.$$

The coefficient λ_{ij}^L measures the strength of joint downside dependence between stocks i and j . A higher value indicates that the two stocks are more likely to experience extreme negative returns together. If $\hat{\tau}_{ij} \leq 0$, the pair is treated as having no positive lower-tail dependence and λ_{ij}^L is set to zero.

This measure is central to the dissertation because it directly captures the type of co-movement that matters most during crises. While Pearson correlation asks whether two stocks move together on average, lower-tail dependence asks whether they fall together when market conditions deteriorate. This distinction allows the study to identify hidden downside linkages that may not be visible under average correlation.

3.6 Network Construction Using Planar Maximally Filtered Graphs

The pairwise dependence estimates form a large dependence matrix. With 193 stocks, the number of possible stock-pair relationships is too large to interpret directly. Network filtering is therefore used to extract the most informative links and represent the market as a clearer structure of connected stocks.

The study uses the Planar Maximally Filtered Graph (PMFG). The PMFG retains more information than a Minimum Spanning Tree while still producing a sparse and interpretable network. For a network with N stocks, the PMFG contains:

$$|E| = 3(N - 2)$$

edges, where E denotes the set of retained links.

The PMFG is constructed separately for the correlation and lower-tail dependence matrices. In the correlation framework, stock pairs are sorted by increasing distance, so that the strongest average co-movement links are considered first. In the lower-tail framework, stock pairs are sorted by decreasing λ_{ij}^L , so that the strongest downside dependence links are considered first. An edge is retained only if adding it keeps the graph planar.

The resulting networks are denoted as $PMFG^{corr}$ and $PMFG^{tail}$. The first represents the structure of average co-movement, while the second represents the structure of joint downside co-movement. This distinction is important for the finance story of the dissertation. If the two networks differ, it suggests that the structure of crash transmission is not the same as the structure of normal market co-movement.

3.7 Centrality Measures and Systemic Importance Index

After constructing the filtered networks, the study measures the systemic importance of each stock using network centrality. The purpose is to move from pairwise dependence to market-wide interpretation. A stock may be systemically important not

only because it is volatile or large, but because it occupies a central position in the network through which downside shocks are transmitted.

Four weighted centrality measures are used: weighted degree, betweenness, closeness, and eigenvector centrality. Weighted degree captures the strength of a stock's direct connections. Betweenness captures whether a stock acts as a bridge between different groups of stocks. Closeness captures how easily a stock can reach the rest of the network. Eigenvector centrality captures whether a stock is connected to other important stocks.

These measures capture different dimensions of systemic relevance. For example, a stock with high weighted degree may be directly exposed to many other stocks, while a stock with high betweenness may connect different sectoral clusters. A stock with high eigenvector centrality may be important because it is connected to other systemically relevant stocks. Therefore, systemic importance cannot be reduced to a single type of centrality.

To combine these measures, the study constructs a Systemic Importance Index using principal component analysis. Before applying PCA, all centrality measures are normalized to the interval [0,1]. The index for stock i is defined as:

$$SII_i = w_D C_D^i + w_B C_B^i + w_C C_C^i + w_E C_E^i,$$

where C_D^i , C_B^i , C_C^i , and C_E^i denote weighted degree, betweenness, closeness, and eigenvector centrality, respectively. The weights are obtained from the first principal component.

Two versions of the index are constructed:

$$SII_i^{corr}$$

from the Pearson correlation network, and:

$$SII_i^{tail}$$

from the Clayton lower-tail dependence network.

The tail-based index is the main systemic importance measure in this study. A stock with high SII^{tail} is not simply a volatile stock. It is a stock that is strongly embedded in the lower-tail dependence network and may therefore contribute to market-wide downside transmission. This interpretation is important for the dissertation because it links the network result to the broader economic question of which stocks and sectors matter most when market stress becomes systemic.

3.8 Sector-Level Systemic Risk Measurement

Because the study focuses on sectoral systemic risk, stock-level systemic importance is aggregated to the sector level. For each sector s , the sector-level SII is computed as:

$$SII_s = \frac{1}{N_s} \sum_{i \in s} SII_i,$$

where N_s is the number of stocks in sector s .

This aggregation allows the analysis to move from individual stock rankings to sectoral risk interpretation. This is necessary because the economic channels of systemic risk in Vietnam are often sector-based. Financials and Real Estate may be connected through credit exposure, collateral values, refinancing pressure, and investor sentiment. Materials and Energy may become important during commodity, construction, or trade-related shocks. Consumer and industrial sectors may transmit stress through domestic demand, export exposure, and supply-chain conditions.

The sector-level measure therefore helps identify whether downside-risk transmission is concentrated in a persistent systemic core or whether it changes across crisis regimes. This is directly connected to the main story of the dissertation: systemic risk in Vietnam is not evenly distributed across the market, but organized through specific stock and sectoral channels.

3.9 Regime-Based Analysis

The analysis is conducted for the full sample and for selected market regimes. The purpose is to examine whether lower-tail dependence and systemic importance are stable over time or change according to the nature of the shock. This is important because systemic risk is state-dependent. A stock or sector may appear less central in normal periods but become important when a specific stress channel becomes active.

For each regime, the study re-estimates dependence, reconstructs the correlation and lower-tail PMFG networks, recalculates centrality measures, and recomputes SII^{corr} and SII^{tail} . This produces comparable stock-level and sector-level systemic importance rankings across market conditions.

The regime-based design allows the study to distinguish between two types of systemic importance. Persistent systemic sectors are sectors that remain central across most regimes, suggesting structural importance in the Vietnamese market. Crisis-specific systemic sectors are sectors whose importance rises sharply during particular shocks, suggesting exposure to specific transmission channels such as trade disruption, pandemic stress, domestic funding pressure, or tariff-related uncertainty.

This step is essential for the economic interpretation of the results. It allows the dissertation to show not only which sectors are systemically important, but also why their importance may change across different crisis environments.

3.10 Framework Comparison and Crisis Validation

The final stage evaluates whether the lower-tail framework provides information beyond conventional correlation. This is done through network comparison, ranking comparison, and crisis validation.

First, the study compares the edge overlap between $PMFG^{corr}$ and $PMFG^{tail}$ using the Jaccard similarity index:

$$J(E^{corr}, E^{tail}) = \frac{|E^{corr} \cap E^{tail}|}{|E^{corr} \cup E^{tail}|}.$$

A low Jaccard value indicates that the correlation network and the lower-tail network retain different links. Economically, this would suggest that the market's average co-movement structure is not the same as its crash co-movement structure.

Second, the study compares SII^{corr} and SII^{tail} using Spearman rank correlation. This tests whether the two frameworks rank systemic stocks similarly. Stocks with large differences between the two rankings are interpreted as cases where conventional correlation may understate or overstate crisis-relevant systemic importance.

Third, the study estimates a cross-sectional crisis validation regression:

$$\bar{R}_i^{crisis} = \alpha + \beta_{tail} SII_i^{tail} + \beta_{corr} SII_i^{corr} + \varepsilon_i,$$

where $\bar{R}_i^{crisis}$ is the average daily return of stock i during a crisis window. Both SII_i^{tail} and SII_i^{corr} are standardized before estimation, and heteroskedasticity-robust standard errors are used.

The key coefficient is β_{tail} . If β_{tail} is negative and statistically significant, this indicates that stocks with higher tail-based systemic importance suffer larger losses during crisis periods. This would support the economic relevance of the lower-tail network. If β_{corr} is weaker or insignificant, it would suggest that lower-tail dependence provides more useful information about crisis vulnerability than conventional correlation.

This validation step is important because the purpose of the study is not only to construct a network, but to test whether the network captures economically meaningful downside risk. A systemic importance measure is useful only if it helps explain which stocks are more vulnerable when the market enters a stress state.

CHAPTER 4 EMPIRICAL RESULTS

4.1 From Market Composition to Tail-Risk Motivation

This subsection describes the empirical sample and explains why a lower-tail framework is appropriate for analysing systemic downside risk on HOSE. The purpose is not only to report the sectoral composition of the data, but also to show that the Vietnamese equity market has structural and distributional features that make crash co-movement economically relevant.

Table 2: Sample composition by GICS sector reports the sectoral composition of the final sample. After applying the trading-history, data-availability and size-liquidity filters, the final dataset consists of 193 HOSE-listed stocks across 11 GICS sectors. Real Estate, Consumer Staples and Financials are the three largest sectors by stock count, accounting for 21.24%, 18.13% and 15.54% of the sample, respectively. Industrials, Materials and Utilities also account for meaningful shares of the sample, while Information Technology, Energy, Health Care and Media Services are more narrowly represented. This structure provides a broad cross-section of the Vietnamese equity market while retaining stocks with sufficient trading continuity for lower-tail dependence estimation.

Table 2: Sample composition by GICS sector

Sector	Number of stocks	Percentage
Real Estate	41	21.24%
Consumer Staples	35	18.13%
Financials	30	15.54%
Industrials	26	13.47%
Materials	19	9.84%
Utilities	18	9.33%
Consumer Discretionary	12	6.22%
Information Technology	5	2.59%
Energy	4	2.07%
Health Care	2	1.04%
Media Services	1	0.52%
Total	193	100.00%

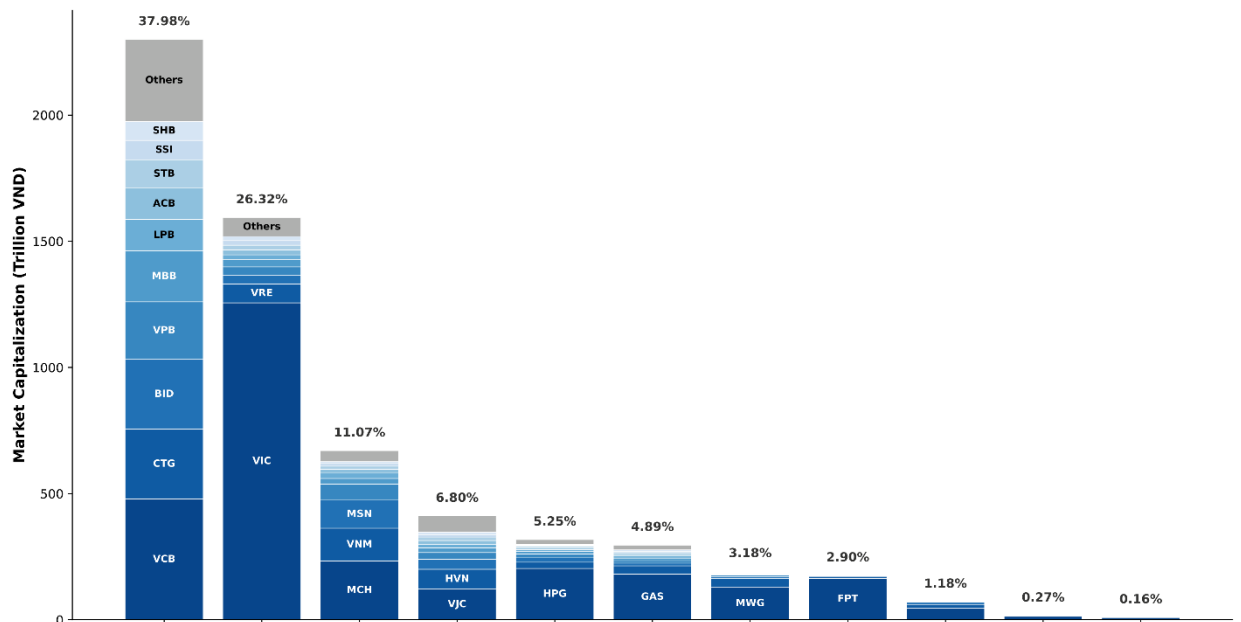
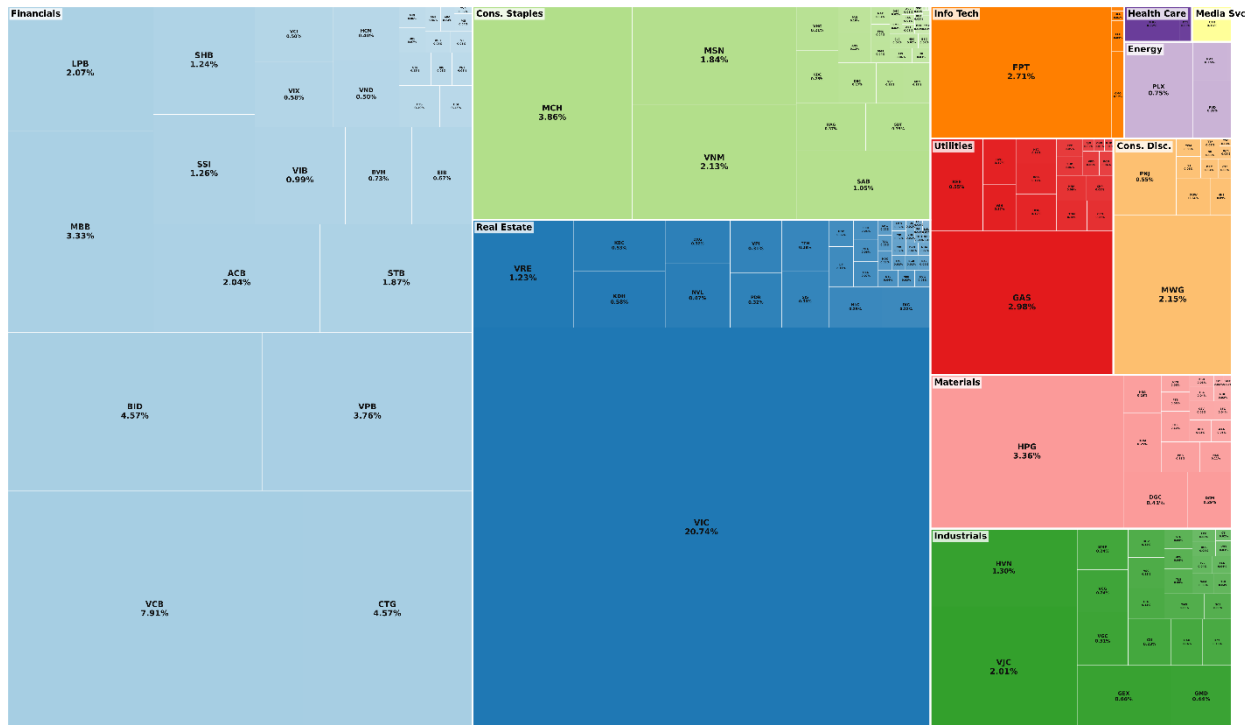


Figure 1: Overall sectoral market capitalization structure

Figure 1 show that the sectoral market-capitalization structure is more concentrated than the stock-count distribution. Financials, Real Estate and Consumer Staples account

for 37.98%, 26.32% and 11.07% of total sectoral market capitalization, respectively. This concentration is important for the interpretation of systemic risk. In Vietnam, Financials and Real Estate are not only large sectors, but also economically connected through credit provision, collateral values, refinancing conditions, corporate bond exposure, margin lending and investor sentiment. Stress in one of these sectors may therefore affect the other and may spread further to materials, construction, consumer and liquidity-sensitive stocks. The sample composition is therefore suitable for examining whether downside risk is transmitted through sectoral channels rather than remaining isolated at the individual-stock level.

The descriptive statistics in Table 3 provide further motivation for using a tail-risk framework. Average daily returns are positive across all sectors, but the study is concerned mainly with downside-risk transmission rather than mean return performance. The volatility pattern shows substantial cross-sectoral heterogeneity. Real Estate records the highest standard deviation, at 2.7390%, followed by Consumer Discretionary at 2.5909% and Consumer Staples at 2.5359%. Financials and Industrials also display relatively high volatility, with standard deviations of 2.4257% and 2.4363%, respectively. By contrast, Utilities has the lowest standard deviation, at 1.8912%. These differences indicate that sectoral risk on HOSE is not uniform. Some sectors are more exposed to liquidity conditions, speculative trading, credit cycles, commodity shocks or domestic demand conditions than others.

Table 3: Descriptive statistics of daily log returns by GICS sector

Sector	Mean (%)	Std. dev. (%)	Min (%)	Max (%)	Skewness	Exc. kurtosis	JB
Consumer Discretionary	0.065	2.591	-7.246	7.317	0.090	1.374	2,368
Consumer Staples	0.054	2.536	-7.459	7.500	0.109	1.722	10,822
Energy	0.052	2.340	-7.311	7.439	-0.021	1.678	1,081
Financials	0.084	2.426	-7.500	7.495	0.071	1.634	8,115
Health Care	0.080	2.122	-7.000	7.006	0.493	2.721	1,740
Industrials	0.068	2.436	-7.478	7.490	0.119	1.654	7,363
Information Technology	0.073	2.483	-7.111	7.143	0.173	1.478	1,197
Materials	0.078	2.376	-7.493	7.407	0.057	1.610	5,096
Media Services	0.115	2.387	-7.372	7.393	0.089	1.469	182
Real Estate	0.072	2.739	-7.435	7.492	0.094	1.133	5,507
Utilities	0.063	1.891	-7.450	7.495	0.161	3.535	22,560

The higher-order moments also show that the return distributions depart from normality. All sectors report positive excess kurtosis, indicating fat-tailed return behaviour. Utilities records the highest excess kurtosis, at 3.5345, followed by Health Care at 2.7212, Consumer Staples at 1.7224, Energy at 1.6778 and Industrials at 1.6544. The Jarque–Bera statistics are large across all sectors, indicating that non-normality is a market-wide feature rather than a sector-specific anomaly. This is important because systemic risk is mainly associated with extreme downside states, where standard average-dependence measures may be less informative. The presence of fat tails and non-normality therefore supports the use of a lower-tail dependence framework.

Table 4 reports the median GJR-GARCH parameters by sector. The estimates show strong volatility persistence across the market, with persistence close to unity in most sectors. Materials, Real Estate, Information Technology, Consumer Staples and Financials record persistence values of 0.9950, 0.9940, 0.9910, 0.9908 and 0.9891, respectively. This indicates that volatility shocks in HOSE-listed stocks tend to decay slowly. The persistence of volatility is relevant for the empirical design because dependence estimated from raw returns may partly reflect time-varying volatility rather than genuine cross-stock downside connectedness.

Table 4: GJR-GARCH persistence and asymmetric volatility response by GICS sector

Sector	Median α	Median γ	Median β	Persistence	$\alpha+\gamma$	$(\alpha+\gamma)/\alpha$
Consumer Discretionary	0.1355	0.0306	0.8141	0.9649	0.1661	1.26
Consumer Staples	0.2328	0.0685	0.7237	0.9908	0.3014	1.45
Energy	0.1015	0.0879	0.8426	0.9880	0.1894	1.52
Financials	0.1368	0.0768	0.8139	0.9891	0.2136	1.44
Health Care	0.2702	0.0574	0.7011	1.0000	0.3276	1.25
Industrials	0.1358	0.0536	0.8245	0.9871	0.1894	1.33
Information Technology	0.2166	0.0334	0.7577	0.9910	0.2500	1.22
Materials	0.1087	0.0813	0.8457	0.9950	0.1900	1.79
Media Services	0.1224	0.2545	0.7503	1.0000	0.3769	3.08
Real Estate	0.1577	0.0487	0.8119	0.9940	0.2064	1.25
Utilities	0.2237	0.0478	0.7576	1.0052	0.2715	1.15

The asymmetric volatility response provides additional evidence that downside shocks matter more than average fluctuations. Among the broader sectors, Materials has the highest negative-to-positive volatility response ratio, at 1.79, followed by Energy at

1.52, Consumer Staples at 1.45 and Financials at 1.44. These values indicate that negative shocks generate stronger conditional-volatility responses than positive shocks of comparable size. This pattern is economically meaningful. Materials and Energy are exposed to commodity prices, construction demand, input costs and investment-cycle conditions. Consumer Staples in Vietnam includes not only stable large-cap consumer firms, but also agricultural, aquaculture, food-processing and export-oriented firms that are sensitive to input costs, export demand and working-capital conditions. Financials are exposed to credit conditions, market liquidity, margin lending, asset quality and investor confidence.

Real Estate records a lower asymmetry ratio of 1.25, but this should not be interpreted as limited systemic relevance. The sector has the highest unconditional volatility and remains structurally connected to the financial system through bank credit, refinancing conditions, collateral values and policy-sensitive project implementation. In stress periods, property-sector weakness may therefore transmit to financial stocks through credit exposure, collateral deterioration, bond-market confidence and margin-related liquidity pressure. This is precisely the type of sectoral mechanism that motivates the lower-tail network analysis in the following sections.

Overall, the preliminary evidence supports the empirical strategy of the dissertation. The market-capitalization structure shows that HOSE is concentrated around sectors with clear systemic relevance, especially Financials and Real Estate. The descriptive statistics show that sectoral returns are volatile, fat-tailed and non-normal. The GJR-GARCH diagnostics show that volatility is persistent and often responds more strongly to negative shocks. These features indicate that systemic risk on HOSE should not be examined only through average returns or unconditional volatility. Instead, the market structure and return behaviour both motivate a lower-tail approach that focuses on whether downside losses become broad-based, synchronized and concentrated around specific sectoral channels.

4.2 Regime-Dependent Lower-Tail Dependence

This subsection examines whether lower-tail dependence among HOSE-listed stocks changes across market regimes. The purpose is to assess whether downside co-movement is a stable feature of the market or whether it intensifies when stress becomes more concentrated. To guide the interpretation, the analysis uses market breadth as the main crisis-context indicator, while the VN-Index is retained as a supporting measure of aggregate market decline. This distinction is important because the index shows how far the market falls, whereas market breadth shows how widely weakness spreads across stocks, thereby mitigating market-capitalisation bias. Lower-tail dependence adds a further dimension by measuring whether these stocks become jointly connected in extreme downside states.

Table 5: Summary of Clayton lower-tail dependence coefficients

Regime	Mean λ_L	Median λ_L	90th pct.	Std. dev.
Full Sample	0.1273	0.0994	0.2991	0.1230
Normal	0.1418	0.1170	0.3264	0.1320
Trade War	0.1058	0.0341	0.3238	0.1435
COVID-19	0.2103	0.1611	0.5185	0.2048
Bond Shock	0.3815	0.4156	0.6992	0.2533
Trump Tariff	0.6106	0.6943	0.8874	0.2776

Table 5 reports the estimated Clayton lower-tail dependence coefficients for the full sample and for each market regime. In the full sample, the mean and median lower-tail dependence coefficients are 0.1273 and 0.0994, respectively, while the 90th percentile reaches 0.2991. These values indicate that downside connectedness exists among HOSE-listed stocks, but the typical pairwise lower-tail dependence remains moderate when the entire 2016–2025 period is pooled together. This is expected because the full sample combines normal trading periods, gradual corrections, sharp crisis episodes and recovery phases. Therefore, the full-sample estimate should be interpreted as a baseline level of latent downside connectedness rather than as a complete measure of systemic crash risk.

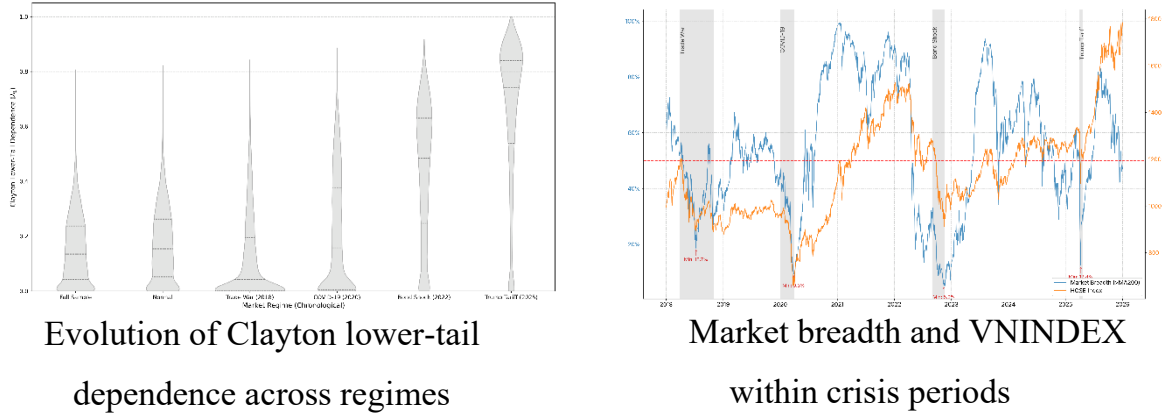


Figure 2: Lower-tail dependence and market breadth across crisis regimes

Figure 2 complements the lower-tail dependence estimates by comparing them with market breadth conditions. The contrast between these indicators is central to the interpretation of crisis transmission. The VN-Index captures aggregate market decline, but because it is capitalization-weighted, it can be driven by a limited number of large stocks. Market breadth captures whether weakness is widespread across the stock universe, while lower-tail dependence captures whether downside movements are synchronized in extreme negative states. Therefore, index movements and lower-tail dependence do not necessarily coincide. A large index decline may reflect sharp losses in several large-cap stocks without broad co-crash behaviour across the market. Conversely, a smaller index decline may still occur together with high lower-tail dependence if many stocks fall jointly while large index constituents partly cushion the aggregate index. A crisis can therefore involve a large index decline without strong lower-tail synchronization, or a smaller index decline with more concentrated co-crash behaviour.

The Trade War period illustrates the first case. The mean and median lower-tail dependence coefficients decline to 0.1058 and 0.0341, respectively, below the normal-period values of 0.1418 and 0.1170. This suggests that the 2018 shock did not generate strong joint lower-tail dependence across HOSE-listed stocks. Although the VN-Index fell from 1196.61 to 914.76, equivalent to a decline of 23.55%, market breadth indicates that the correction was uneven. During this window, the share of stocks trading above

their 200-day moving average ranged from 18.65% to 56.48%, with an average of 38.51%. This pattern suggests that downside pressure was present, but it did not translate into a highly synchronized lower-tail structure across the market. Economically, the Trade War appears to have operated more as a gradual external repricing shock than as a broad domestic systemic event.

COVID-19 shows a clearer increase in downside synchronization. The mean lower-tail dependence coefficient rises to 0.2103, while the median increases to 0.1611 and the 90th percentile reaches 0.5185. Market breadth also deteriorates more severely than during the Trade War, with the share of stocks above their 200-day moving average falling to a minimum of 9.33% and averaging 30.54%. This indicates that the pandemic shock produced a broader deterioration in market participation. The increase in lower-tail dependence is consistent with the nature of COVID-19 as a simultaneous shock to domestic mobility, consumption, production activity, credit expectations, liquidity and investor sentiment. In this regime, downside risk became more synchronized because several sectors were exposed to the same adverse macroeconomic and financial conditions.

The Bond Shock regime provides the strongest evidence that market breadth can reveal systemic stress more clearly than the index alone. The mean lower-tail dependence coefficient increases to 0.3815, the median rises to 0.4156 and the 90th percentile reaches 0.6992. Although the VN-Index decline of 26.18% is smaller than the COVID-19 drawdown, market breadth is much weaker. The share of stocks above their 200-day moving average falls to a minimum of 5.18% and averages only 14.81%, the weakest breadth condition among the four crisis periods. This suggests that the Bond Shock was not only a market correction, but also a broad-based deterioration in stock participation and trend strength. Economically, this is consistent with a domestic funding shock in which corporate bond stress, refinancing constraints, weakening confidence in leveraged real-estate firms and margin-related selling pressure transmitted downside risk to financial institutions and liquidity-sensitive stocks.

The Trump Tariff episode records the highest lower-tail dependence across all regimes. The mean coefficient reaches 0.6106, the median increases to 0.6943 and the 90th percentile rises to 0.8874. This result is notable because the VN-Index decline during the event window is only 7.46%, from 1317.33 to 1219.12. The high lower-tail dependence therefore cannot be explained by the size of the index drawdown alone. Instead, it indicates a rapid synchronization of downside movements across stocks. Market breadth supports this interpretation. The share of stocks above their 200-day moving average drops to a minimum of 12.44% and averages 29.29% within a short event window. This suggests that the tariff shock operated as a sudden external repricing event, triggering a cross-sector risk-off response and a quick deterioration in market participation.

Taken together, the regime-level evidence shows that lower-tail dependence is strongly state-dependent. The Trade War involved a sizeable index correction, but lower-tail dependence remained comparatively weak, suggesting that the shock was absorbed unevenly across stocks. COVID-19 generated a stronger increase in joint downside dependence because the shock affected both financial conditions and real economic activity. The Bond Shock produced the sharpest deterioration in market breadth and a much higher median λ_L , consistent with a domestic credit, collateral and refinancing channel. The Trump Tariff episode generated the highest lower-tail dependence despite the smallest index decline, indicating that sudden external repricing shocks can produce highly synchronized downside movements even over short windows.

These findings support the use of a lower-tail dependence framework for systemic-risk analysis on HOSE. Standard market indicators capture only part of the crisis story. The VN-Index measures aggregate price decline, while market breadth measures the extent of participation deterioration. Clayton lower-tail dependence adds the systemic dimension by showing whether downside movements become jointly concentrated across stocks. Therefore, systemic downside risk in Vietnam is not determined only by

how far the market falls, but also by how broadly and synchronously weakness spreads through the equity network.

4.3 Sectoral Channels of Downside Co-movement

This subsection examines whether lower-tail dependence is randomly distributed across stock pairs or organized around economically meaningful sectoral channels. The results show that downside co-movement is concentrated in both intra-sector clusters and cross-sector transmission links. This distinction is important because systemic downside risk may arise from two sources: common exposures within the same sector and spillover channels across sectors. The analysis therefore first examines intra-sector lower-tail clusters, then discusses how these clusters change across regimes, and finally identifies the main cross-sector channels of crash transmission.

4.3.1 Intra-sector Lower-tail Clusters

Table 6 reports the strongest intra-sector lower-tail dependent pairs across the full sample and crisis regimes. In the full sample, the leading pairs are HCM–SSI, HSG–NKG, SSI–VND, ASM–IDI and DCM–DPM, with lower-tail dependence coefficients of 0.782, 0.757, 0.752, 0.739 and 0.716, respectively. These values are substantially higher than the full-sample median lower-tail dependence reported in Table 3, indicating that the strongest downside linkages are concentrated within specific sectoral clusters. Financials account for six of the top ten intra-sector pairs, while Materials and Consumer Staples also appear among the strongest lower-tail relationships.

The concentration of Financials in the intra-sector ranking is consistent with the structure of Vietnam’s financial market. Securities-firm pairs such as HCM–SSI, SSI–VND, HCM–VCI, HCM–VND and SSI–VCI form a persistent lower-tail cluster because these firms share exposure to market turnover, margin lending, proprietary trading, capital-market activity and investor risk appetite. During market downturns, declining liquidity, weaker brokerage income, lower margin demand, mark-to-market losses and

forced deleveraging can occur simultaneously, leading to joint downside movements across securities firms.

Table 6: Top intra-sector lower-tail dependent pairs

Pair	Sector	Full Sample	Normal	Trade War	COVID-19	Bond Shock	Trump Tariff
HCM-SSI	Financials	0.782	0.805	0.770	0.829	0.804	0.914
HSG-NKG	Materials	0.757	0.796	0.565	0.657	0.899	0.927
SSI-VND	Financials	0.752	0.765	0.781	0.687	0.865	0.961
ASM-IDI	Consumer Staples	0.739	0.730	0.700	0.813	0.835	0.914
DCM-DPM	Materials	0.716	0.774	0.494	0.645	0.899	0.873
ACB-MBB	Financials	0.708	0.703	0.819	0.840	0.736	0.927
HCM-VCI	Financials	0.699	0.738	0.577	0.642	0.860	0.873
HCM-VND	Financials	0.697	0.719	0.704	0.670	0.747	0.857
CTG-MBB	Financials	0.696	0.693	0.831	0.719	0.841	0.873
SSI-VCI	Financials	0.692	0.733	0.612	0.497	0.845	0.805

Banking pairs such as ACB–MBB and CTG–MBB also show strong lower-tail dependence. This result should not be interpreted as evidence that these banks have identical credit structures. Rather, it reflects a system-wide banking channel. During stress periods, investors may reprice banks together because of common concerns over funding costs, interbank liquidity, credit-growth constraints, asset quality and confidence toward the banking system.

Materials form the second important intra-sector channel. The steel pair HSG–NKG and the fertilizer pair DCM–DPM reflect commodity-linked dependence. HSG and NKG share exposure to steel prices, input costs, construction demand, export orders and trade-policy conditions. DCM and DPM are linked through urea prices, gas input costs, agricultural demand and regional supply conditions. These results show that lower-tail dependence reflects real-sector fundamentals as well as financial-market linkages.

The Consumer Staples pair ASM–IDI shows that defensive-sector labels should be interpreted carefully in Vietnam. Although Consumer Staples is often viewed as defensive, ASM and IDI are connected through aquaculture, seafood processing, export demand and group-level business exposure. Shocks to fish prices, input costs, working-capital conditions or export markets can therefore affect both stocks simultaneously.

4.3.2 Regime-specific Shifts in Intra-sector Dependence

Figure 3 illustrates how intra-sector downside dependence changes across regimes. During the Trade War period, the strongest pairs are concentrated mainly in Financials, especially among banks and securities firms. This suggests that the external trade shock



Figure 3: Strongest intra-sector lower-tail dependent pairs

was partly transmitted through investor reassessment of domestic liquidity, financial-sector confidence and funding conditions. However, compared with later crisis episodes, the lower-tail structure remained less synchronized, consistent with the interpretation that the Trade War shock was more gradual and heterogeneous across firms.

During COVID-19, the structure becomes more diversified, combining Financials with consumer-facing and operating-activity-sensitive firms. This is consistent with the pandemic shock, which affected not only financial conditions but also consumption, mobility, store operations, production activity and supply chains. The presence of pairs such as MWG–PNJ and ASM–IDI suggests that lower-tail dependence during this period was linked not only to market liquidity, but also to disruptions in real economic activity.

The Bond Shock shows a different pattern. The strongest intra-sector pairs span Real Estate, Materials, Industrials and Financials, suggesting that domestic funding stress affected several credit-sensitive and liquidity-sensitive sectors at the same time. Real Estate pairs such as LDG–SCR reflect the vulnerability of leveraged property stocks to refinancing pressure, corporate bond-market stress, margin-related selling and confidence deterioration. The appearance of Materials and Industrials also suggests that domestic funding stress was not confined to property developers, but was connected to construction demand, infrastructure expectations and broader real-economy activity.

The Trump Tariff episode is dominated by very high lower-tail dependence among Real Estate and other high-beta pairs. Several Real Estate pairs record lower-tail dependence coefficients close to or above 0.90. This suggests that the tariff shock triggered rapid cross-stock repricing, especially among stocks perceived as sensitive to liquidity withdrawal, external uncertainty and risk-off behaviour. This pattern is consistent with the earlier evidence that a relatively small index decline can still coincide with strong lower-tail synchronization when downside movements become concentrated across many stocks.

Overall, the regime-specific evidence shows that intra-sector lower-tail dependence is not stable over time. Financials remain a recurrent source of downside connectedness, but the dominant sectoral clusters change according to the nature of each shock. External trade shocks are reflected mainly through Financials and high-beta repricing, pandemic stress broadens dependence across financial and real-economy sectors, domestic funding stress strengthens Real Estate and credit-sensitive clusters, and tariff-related uncertainty increases co-movement among externally sensitive and liquidity-sensitive stocks.

4.3.3 Cross-sector Transmission Channels

Table 7 reports the strongest cross-sector lower-tail dependent pairs. These pairs are especially important because they show how downside risk can move across sector

boundaries. In the full sample, the strongest cross-sector pairs include HHS–TCH, DLG–HQC, GAS–PVD, HPG–SSI, GAS–PLX, DXG–SSI, KSB–LCG, DXG–VND, DXG–HCM and HCM–HPG. These links point to several transmission channels: Real Estate–Financials, Materials–Financials, Utilities–Energy, Materials–Industrials and high-beta cross-sector sentiment channels.

The Real Estate–Financials channel is the most important cross-sector transmission mechanism. Pairs such as DXG–SSI, DXG–VND and DXG–HCM connect property developers with securities firms. This channel combines credit-collateral risk with market-liquidity risk. DXG is sensitive to housing-market liquidity, refinancing conditions, legal progress, investor sentiment and asset-price expectations. SSI, VND and HCM are exposed to trading value, margin lending, proprietary investment and capital-market sentiment. When real-estate stress rises, declining collateral values, margin pressure, forced selling and weaker market turnover can affect both property and securities stocks simultaneously.

Table 7: Top cross-sector lower-tail dependent pairs

Pair	Sector 1	Sector 2	Full Sample	Normal	Trade War	COVID-19	Bond Shock	Trump Tariff
HHS-TCH	Consumer Discretionary	Real Estate	0.606	0.682	0.142	0.501	0.890	0.857
DLG-HQC	Information Technology	Real Estate	0.576	0.624	0.318	0.642	0.918	0.927
GAS-PVD	Utilities	Energy	0.575	0.572	0.640	0.707	0.375	0.950
HPG-SSI	Materials	Financials	0.546	0.556	0.649	0.641	0.595	0.927
GAS-PLX	Utilities	Energy	0.546	0.523	0.685	0.536	0.494	0.914
DXG-SSI	Real Estate	Financials	0.539	0.496	0.693	0.668	0.770	0.972
KSB-LCG	Materials	Industrials	0.527	0.586	0.013	0.517	0.819	0.901
DXG-VND	Real Estate	Financials	0.524	0.505	0.601	0.506	0.783	0.927
DXG-HCM	Real Estate	Financials	0.523	0.501	0.614	0.764	0.756	0.927
HCM-HPG	Financials	Materials	0.522	0.527	0.574	0.561	0.576	0.927

The Materials–Financials channel appears through HPG–SSI and HCM–HPG. HPG is linked to steel prices, construction demand, infrastructure expectations and the property cycle, while securities firms are sensitive to market liquidity and margin activity. This suggests that real-economy cyclicalities and financial-market leverage may reinforce each other during downside episodes. When construction or steel demand weakens,

investors may also reduce exposure to market-sensitive financial intermediaries, creating joint lower-tail movements between Materials and Financials.

The Utilities–Energy channel, represented by GAS–PVD and GAS–PLX, reflects common exposure to energy-sector sentiment, oil and gas price expectations, inflation risk and broad risk-off selling in energy-related stocks. GAS is more closely linked to gas infrastructure and midstream activities, PVD is exposed to drilling and upstream service demand, while PLX reflects petroleum distribution and fuel-market conditions. These links are therefore better interpreted as energy-theme repricing channels rather than direct firm-to-firm earnings linkages.

The Materials–Industrials pair KSB–LCG captures another real-economy transmission channel. KSB is exposed to construction materials and infrastructure demand, while LCG is linked to construction and contractor-related activity. Their lower-tail dependence suggests that shocks to infrastructure expectations, construction demand or domestic investment cycles can generate cross-sector downside co-movement between suppliers and contractors.

Taken together, the sectoral evidence shows that lower-tail dependence on HOSE is economically organized. Financials form the most persistent intra-sector cluster, especially through securities firms and selected banks. Materials display strong commodity-cycle dependence through steel and fertilizer pairs. Real Estate is central to cross-sector downside transmission through its links with Financials and other high-beta stocks. Energy and Utilities reflect commodity and energy-theme repricing, while selected Consumer Staples pairs show that export-oriented and agriculture-linked firms can behave cyclically during stress periods.

These findings answer the sectoral dimension of RQ1. Lower-tail dependence among HOSE-listed stocks is not only present, but also structured around identifiable channels of crash transmission. The Financials–Real Estate channel reflects credit, collateral, refinancing, margin lending and market-liquidity mechanisms. The Materials–

Energy–Industrials channel reflects commodity prices, construction demand, input costs and investment-cycle exposure. Selected Consumer Staples and high-beta cross-sector pairs reflect export demand, working-capital risk and investor sentiment. Therefore, systemic downside risk in Vietnam is better understood as a sectorally organized network of lower-tail co-movement rather than as a uniform market-wide decline.

4.4 Systemic Importance and Crash Transmission across Regimes

This subsection examines which stocks and sectors become most important in the lower-tail dependence network. While the previous subsection focuses on pairwise downside co-movement, the present analysis moves from bilateral dependence to systemic importance. This distinction is important because a stock may be strongly linked to a few similar firms without being central to market-wide crash transmission. Conversely, a stock may become systemically important when it connects several parts of the lower-tail network or occupies a central position in the structure of co-crash dependence.

Two complementary indicators are used. The first is the Clayton-based systemic importance index, SII_{tail} , which measures each stock's centrality within the lower-tail PMFG network. A high SII_{tail} indicates that the stock is structurally embedded in the downside transmission network. The second is hub frequency, defined as the number of times a stock appears in the top 10% of pairwise lower-tail dependence links. While SII_{tail} captures network position, hub frequency captures repeated participation in strong co-crash relationships. These measures should not be interpreted simply as indicators of firm size, standalone volatility or firm quality. Rather, they indicate how the market historically priced certain stocks as focal points of downside-risk transmission across different regimes.

4.4.1 Tail-based Systemic Importance across Regimes

Figure 4 reports the leading SII_{tail} stocks across the full sample and market regimes. In the full sample, systemic importance is concentrated around Financials and

Real Estate. The leading stocks include SSI, DXG, HCM, MBB, DIG, SCR, CTS, HPG, AGR and IJC. This ranking indicates that the lower-tail network is anchored by securities firms, banks, property developers and selected cyclical or infrastructure-related stocks. The result is consistent with the structure of Vietnam's equity market, where Financials and Real Estate are closely connected through credit provision, collateral values, refinancing conditions, corporate bond exposure, margin lending and investor sentiment.

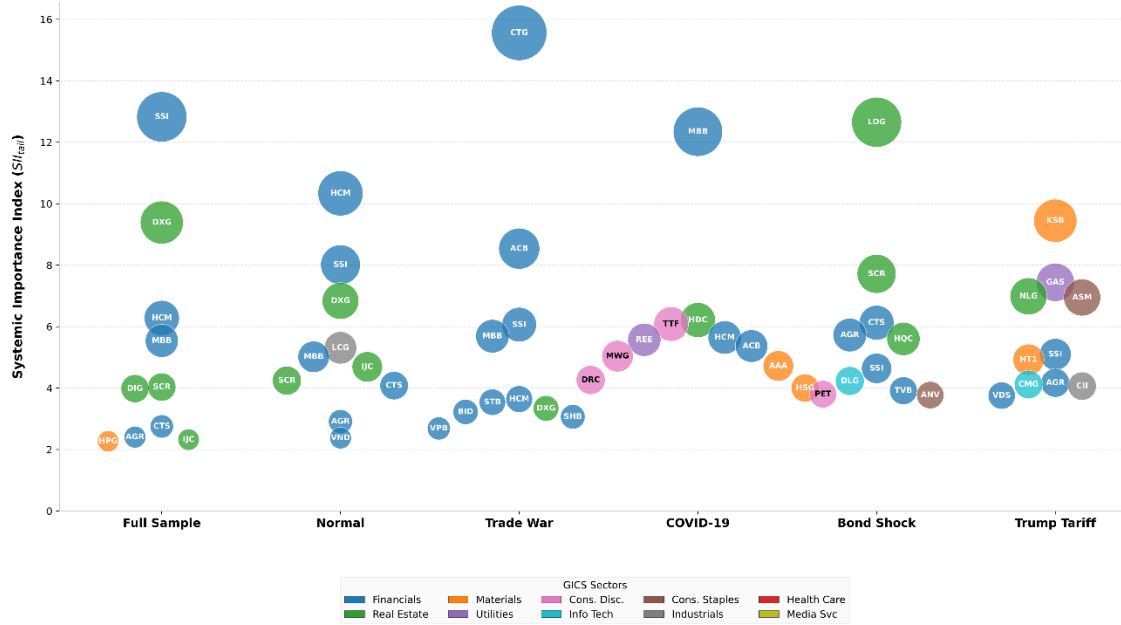


Figure 4: Top SII_{tail} stocks across market regimes stocks

The full-sample ranking suggests that the Financials–Real Estate nexus is not only a crisis-specific phenomenon, but a persistent feature of the downside-risk network. Securities firms such as SSI and HCM are central because their earnings and risk exposures are directly linked to trading turnover, margin financing, proprietary investment and market liquidity. Banks such as MBB are important because they represent credit conditions, deposit competition, asset-quality expectations and financial-system confidence. Real-estate stocks such as DXG, DIG and SCR are central because their valuations are sensitive to refinancing conditions, project liquidity, collateral values and investor expectations toward property assets. When market stress

emerges, these sectors can reinforce one another through credit tightening, collateral deterioration, margin pressure and risk-averse portfolio rebalancing.

The regime-specific rankings show that systemic importance is not fixed over time. During the Trade War period, the systemic core remains strongly financial, with CTG, ACB, SSI, MBB, STB, HCM, DXG, BID, SHB and VPB appearing among the leading stocks. This suggests that the external trade shock was partly transmitted through investor reassessment of banks, securities firms and broad financial-sector confidence. Although the Trade War did not generate the strongest lower-tail dependence at the market-wide level, the stocks that became central were still concentrated in sectors associated with liquidity, credit conditions and market sentiment.

During COVID-19, systemic importance becomes more diversified across sectors. The appearance of MBB and HCM reflects the continued importance of credit conditions and securities-market liquidity, while HDC, REE, MWG, AAA, DRC and HSG point to real-estate activity, utilities, consumer-facing businesses, materials and industrial demand. This broader composition is consistent with the nature of the pandemic shock. COVID-19 affected not only financial-market conditions, but also household consumption, store operations, supply chains, production activity and expectations about future cash flows. As a result, the lower-tail network became more cross-sectoral, linking financial stocks with firms exposed to real economic disruption.

The Bond Shock provides the clearest evidence of a domestic credit-collateral transmission regime. Systemic importance becomes concentrated around leveraged real-estate and securities-related stocks, including LDG, SCR, CTS, AGR, HQC, SSI, DLG and TVB. This pattern is consistent with an episode driven by weaker corporate bond-market confidence, refinancing pressure and concerns over real-estate balance sheets. In this regime, property developers, collateral values, funding access and liquidity-sensitive financial intermediaries appear to have been priced as part of a common downside channel. As refinancing risk increased and confidence deteriorated, stress could spread

from property developers to securities firms and then to the broader market through margin pressure, forced selling and declining market turnover.

The Trump Tariff regime displays a different form of systemic rotation. The leading SII_{tail} stocks include PET, ANV, KSB, GAS, NLG, ASM, SSI, HT1, CMG, AGR, CII and VDS. This composition indicates a shift toward materials, energy, real estate, consumer staples, infrastructure-linked firms and financial intermediaries. KSB and HT1 represent construction-material and infrastructure-related cyclicity, GAS reflects energy and commodity sentiment, while NLG and CII capture real-estate and infrastructure sensitivity. ASM and ANV indicate that agriculture, aquaculture and export-linked consumer-staples firms can become systemically relevant when trade-policy uncertainty affects expectations. Financial intermediaries remain present because sudden external repricing can reduce market turnover, increase margin pressure and trigger liquidity-sensitive selling.

Taken together, the SII_{tail} rankings indicate that systemic importance is both persistent and state-dependent. Financials and Real Estate form the stable domestic transmission core because they connect credit provision, collateral values, securities-market liquidity and investor leverage. However, the identity of the most systemically important stocks changes when the source of stress changes. COVID-19 broadens systemic importance across financial, real-estate, consumer, utility and materials-related stocks. The Bond Shock concentrates systemic importance in the Real Estate–Financials nexus through refinancing pressure and collateral-value deterioration. The Trump Tariff episode rotates systemic importance toward externally sensitive and high-beta sectors through trade-policy uncertainty, commodity sentiment and market-wide de-risking.

4.4.2 Systemic Hubs and Repeated Co-crash Participation

Figure 5 provides additional evidence on systemic hubs. Unlike SII_{tail} , which captures structural centrality in the filtered lower-tail network, hub frequency measures how often a stock appears in the top 10% of pairwise lower-tail dependence links. This

distinction is useful because a stock may not always be the most central node in the PMFG, but may still repeatedly participate in strong co-crash relationships. Hub frequency therefore helps identify stocks that are persistently involved in intense downside co-movement.

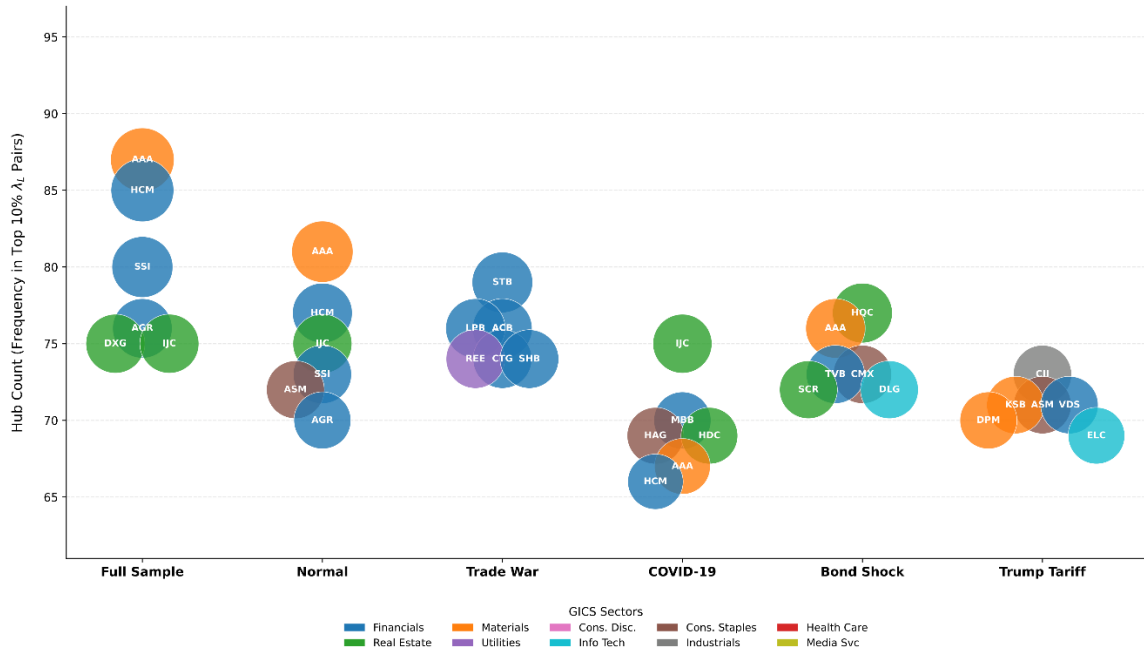


Figure 5: Systemic hub tickers by frequency in top 10% lower-tail dependence pairs

In the full sample and normal periods, recurrent hubs such as AAA, HCM, SSI and IJC suggest that latent downside connectivity was anchored around materials-cycle exposure, securities-market liquidity and real-estate sensitivity. This complements the SII_{tail} results because hub frequency captures repeated pairwise co-crash participation, while SII_{tail} captures structural centrality in the filtered network. The appearance of these names indicates that systemic relevance is not limited to the largest index constituents. Instead, it may also arise from stocks that repeatedly move with other stocks in adverse states.

During the Bond Shock, hub leadership shifts toward stocks associated with real-estate stress, commodity-linked activity and liquidity-sensitive financial transmission. This is consistent with the interpretation that domestic funding pressure activated the credit-collateral channel and increased the importance of property developers, securities

firms and cyclically sensitive stocks. In this regime, repeated co-crash participation reflects the way refinancing concerns, collateral deterioration, margin pressure and liquidity withdrawal became connected across several parts of the market.

During the Trump Tariff episode, hubs such as CII, ASM and KSB indicate that sudden external repricing shifted repeated co-crash linkages toward infrastructure-linked, agriculture-related and construction-material stocks. This pattern reinforces the interpretation that external shocks may rotate systemic relevance away from the domestic Financials–Real Estate core toward sectors exposed to trade uncertainty, commodity sentiment and high-beta market de-risking. Therefore, hub frequency supports the view that systemic importance reflects the market’s perception of active transmission channels rather than firm size alone.

The joint evidence from SII_{tail} and hub frequency provides a direct answer to RQ2. The stocks and sectors contributing most to market-wide downside risk on HOSE are not fixed across time. Financials and Real Estate are the most persistent systemic core because they connect credit intermediation, collateral values, refinancing conditions, margin lending, market liquidity and investor confidence. However, crash transmission does not remain confined to this core. When the dominant shock changes, systemic importance rotates toward the sectors most exposed to that shock. Pandemic stress expands systemic importance across real and financial sectors; domestic funding stress concentrates it in the Real Estate–Financials nexus; and external tariff-related uncertainty shifts it toward Materials, Energy, Consumer Staples, infrastructure-linked stocks and liquidity-sensitive Financials.

Therefore, systemic importance on HOSE should be understood as a regime-dependent network property. A stock becomes systemically important not only because it is large or volatile, but because it occupies a central position in the lower-tail network or repeatedly participates in strong co-crash relationships when a specific transmission channel becomes active. This finding supports the broader argument of the dissertation:

systemic risk in Vietnam's equity market is shaped by the interaction between sectoral structure, investor behaviour, liquidity conditions and the nature of the shock.

4.5 Correlation versus Lower-Tail Dependence

This subsection examines whether the lower-tail framework provides information beyond conventional Pearson correlation. The purpose is not to repeat the conceptual distinction between the two measures, but to test whether this distinction matters empirically. Pearson correlation captures average linear co-movement across the full return distribution, while Clayton lower-tail dependence focuses on joint downside realizations. If the two frameworks identify similar network structures and similar systemic stocks, then the lower-tail approach would add limited information. If they diverge, and if the tail-based measure better explains crisis-period losses, then lower-tail dependence provides economically relevant information for systemic-risk analysis.

Table 8 **Error! Reference source not found.** reports the network divergence and crisis validation results. The Jaccard similarity between the Pearson-based and Clayton-based PMFG networks is low across all crisis regimes. It equals 0.196 during the Trade War, 0.125 during COVID-19, 0.248 during the Bond Shock and 0.130 during the Trump Tariff episode. These values indicate that the correlation and lower-tail networks retain substantially different sets of links. Even in the Bond Shock regime, where the Jaccard value is the highest, more than three quarters of the lower-tail network edges are not shared with the correlation network. This suggests that average co-movement and downside co-crash dependence represent different dimensions of market connectedness.

Table 8: Network Divergence and Crisis Validation

Regime	Jaccard	β_{tail}	p_{tail}	β_{corr}	p_{corr}	R^2
Trade War	0.196	-0.048	0.24317	0.000	0.98792	0.082
COVID-19	0.125	-0.125	0.00484	0.011	0.81306	0.080
Bond Shock	0.248	-0.421	0.01330	0.066	0.51531	0.342
Trump Tariff	0.130	-0.167	0.00005	-0.096	0.01591	0.084

Notes: Jaccard measures the overlap between Pearson and Clayton PMFG edges. Crisis validation regressions relate each stock's average daily log return during the corresponding crisis window to SII_{tail} and SII_{corr} . Both regressors are cross-sectionally standardized. Lower Jaccard values indicate

greater divergence between correlation and lower-tail dependence networks. Negative and significant β_{tail} indicates that stocks with higher lower-tail systemic importance experience larger crash losses.

The low Jaccard similarity is especially important during COVID-19 and the Trump Tariff episode, where the values are only 0.125 and 0.130, respectively. These results imply that most lower-tail connections are not captured by the Pearson-based network. Economically, this means that the market's crash-transmission structure can differ substantially from its normal co-movement structure. During stress periods, dependence may be shaped by liquidity withdrawal, margin deleveraging, forced selling, funding pressure, collateral deterioration and sudden repricing. These mechanisms are more closely related to downside states than to average return co-movement.

The crisis validation regressions provide stronger evidence that the lower-tail framework is economically meaningful. During COVID-19, β_{tail} is negative and statistically significant, with a coefficient of -0.125 and a p -value of 0.00484, while β_{corr} is insignificant. This indicates that stocks with higher lower-tail systemic importance experienced larger losses during the pandemic shock, whereas correlation-based systemic importance did not explain the cross-sectional crisis loss pattern. This result is consistent with COVID-19 as a broad macro-financial shock that affected both real economic activity and market liquidity.

The Bond Shock provides the strongest validation result. The coefficient on SII_{tail} is -0.421 and statistically significant, while SII_{corr} remains insignificant. The regression also records the highest explanatory power, with $R^2 = 0.342$. This suggests that lower-tail centrality was particularly informative during the domestic funding and collateral-stress regime. In this episode, stocks located closer to the lower-tail transmission core suffered larger realized losses. The result is consistent with the interpretation that the Bond Shock activated a credit-collateral channel linking real estate developers, securities firms, funding conditions, collateral values and margin-related selling pressure.

During the Trump Tariff episode, both systemic importance measures are statistically significant, but the lower-tail coefficient is larger in absolute value. The

coefficient on SII_{tail} is -0.167, compared with -0.096 for SII_{corr} . This suggests that the tariff-related shock involved both broad market co-movement and lower-tail co-crash dependence. However, the stronger tail-based coefficient indicates that the Clayton framework still provides the clearer signal of downside vulnerability. This is consistent with the earlier evidence that the Trump Tariff episode produced the highest lower-tail dependence despite a relatively small VN-Index decline.

By contrast, neither SII_{tail} nor SII_{corr} is statistically significant during the Trade War period. This result is consistent with the previous interpretation that the 2018 shock was more gradual and heterogeneous across firms. Although the VN-Index declined substantially, market breadth and lower-tail dependence suggest that downside pressure was not as synchronized as in later crisis episodes. The absence of significant validation coefficients therefore does not weaken the lower-tail argument. Rather, it confirms that the usefulness of systemic importance measures depends on whether a shock generates broad and synchronized downside transmission.

Taken together, the evidence shows that lower-tail dependence adds information beyond Pearson correlation. First, the low Jaccard values confirm that the Clayton and Pearson networks identify different dependence structures. Second, the crisis validation regressions show that SII_{tail} explains realized crisis-period losses more consistently than SII_{corr} . Third, the strongest performance of SII_{tail} occurs in regimes where market stress is associated with funding pressure, collateral deterioration, liquidity withdrawal or rapid external repricing. Therefore, lower-tail dependence is not only technically different from correlation; it provides economically relevant information about which stocks are more vulnerable when downside risk becomes systemic.

These findings answer RQ3. Conventional Pearson correlation remains useful for describing average co-movement, but it does not fully capture the structure of crash transmission on HOSE. The lower-tail framework identifies a different network of downside linkages and provides stronger explanatory power for crisis-period losses. This supports the central argument that systemic risk in Vietnam's equity market is better

understood through a regime-dependent lower-tail network than through average linear co-movement alone.

4.6 Summary of Empirical Findings

This chapter provides three main empirical findings. First, lower-tail dependence is a meaningful and regime-dependent feature of the Vietnamese equity market. The full-sample results show moderate latent downside connectedness, with a mean λ_L of 0.1273 and a median of 0.0994. However, dependence increases sharply during stress regimes, especially during COVID-19, the Bond Shock and the Trump Tariff episode. This indicates that downside co-movement is not constant, but becomes stronger when market stress is broader, faster or more synchronized.

Second, market breadth and sectoral evidence show that systemic downside risk is not captured by index decline alone. The Trade War involved a sizeable index correction but relatively weak lower-tail dependence, while the Bond Shock and Trump Tariff episode showed stronger downside synchronization despite different index movements. This suggests that systemic fragility depends not only on how far the market falls, but also on how widely and jointly weakness spreads across stocks. At the sectoral level, lower-tail dependence is concentrated around identifiable channels, especially the Financials–Real Estate credit-collateral channel and the Materials–Energy–Industrials real-sector channel.

Third, systemic importance is both persistent and regime-dependent. Financials and Real Estate form the stable domestic transmission core of the lower-tail network, with leading SII_{tail} stocks such as SSI, DXG, HCM, MBB, DIG and SCR. However, the identity of systemically important stocks changes across regimes. COVID-19 broadens systemic importance across real and financial sectors, the Bond Shock strengthens the Real Estate–Financials nexus, and the Trump Tariff episode rotates systemic importance toward Materials, Energy, Consumer Staples, infrastructure-linked stocks and liquidity-sensitive Financials.

Finally, the comparison with Pearson correlation confirms that lower-tail dependence provides additional information for crisis-risk analysis. The Pearson and Clayton PMFG networks show low overlap, indicating that average co-movement and downside co-crash dependence identify different market structures. The crisis validation regressions further show that SII_{tail} explains realized crisis-period losses more consistently than SII_{corr} , particularly during COVID-19, the Bond Shock and the Trump Tariff episode.

Taken together, the findings answer the three research questions of the study. HOSE-listed stocks exhibit meaningful lower-tail dependence; this dependence is organized around economically important sectoral channels; and the lower-tail network identifies systemic downside risk more effectively than conventional correlation. Overall, systemic risk in Vietnam's equity market is best understood as a regime-dependent lower-tail network shaped by sectoral linkages, market breadth deterioration and crisis-specific transmission channels.

CHAPTER 5 DISCUSSION AND CONCLUSION

5.1 Discussion of Key Findings

The central argument of this dissertation is that systemic risk in the Vietnamese equity market should be understood as a structure of joint downside vulnerability, rather than as a simple extension of individual stock volatility or average return co-movement. The empirical results support this argument. HOSE-listed stocks exhibit meaningful lower-tail dependence, and this dependence becomes stronger when market stress is associated with liquidity pressure, funding constraints, or rapid external repricing. The increase in lower-tail dependence during the Bond Shock and the U.S. reciprocal tariff shock shows that crisis severity cannot be assessed only by the size of the VN-Index decline. A smaller index drawdown may still involve stronger systemic stress when many stocks move into the lower tail together. This is the main finance story of the study: systemic fragility emerges when losses become synchronized and diversification becomes less reliable precisely when investors need it most (Duy Duong and Duc T. L. Huynh, 2020; Markowitz, 1952; McNeil et al., 2015).

The findings also show that downside co-movement on HOSE is economically structured. The strongest lower-tail links are concentrated around sectors that share credit, collateral, liquidity, commodity, and investment-cycle exposures. Financials and Real Estate form the most persistent transmission core. This result is consistent with the structure of Vietnam's financial system, where banks and securities firms are central to credit provision, margin financing, market liquidity, and investor confidence, while real-estate firms are closely tied to collateral values, refinancing conditions, corporate bond-market confidence, and household wealth expectations. Stress in one side of this nexus can therefore affect the other side through several channels. A decline in property valuations may weaken collateral capacity and investor confidence toward financial institutions. At the same time, tighter financial conditions may reduce real-estate liquidity and reinforce selling pressure in property-related stocks. The lower-tail network captures this mutual dependence more directly than a conventional average co-

movement measure (Anh P. Nguyen et al., 2023; Anh T. H. Nguyen and Thanh T. Le, 2025; Van T. T. Vu and Kham D. Tran, 2019).

Beyond the Financials and Real Estate core, the results identify a second layer of systemic transmission through Materials, Energy, Utilities, and selected industrial or consumer-related stocks. This layer is important because it shows that systemic risk in Vietnam is not purely a financial-sector issue. Materials stocks are exposed to construction demand, real-estate activity, commodity prices, input costs, and export conditions. Energy and Utilities are linked to oil and gas sentiment, infrastructure demand, inflation expectations, and policy-sensitive investment cycles. During external repricing shocks, these sectors can become more central because investors reassess future cash flows, trade exposure, and cyclical demand at the same time. The evidence therefore suggests that crash transmission on HOSE combines financial channels and real-sector channels. This supports the broader view that market-based systemic risk can arise from common exposures and investor revaluation, even without direct contractual linkages across firms (Mensi et al., 2024; Nhat T. H. Dang et al., 2023; Wang et al., 2022).

A further contribution of the study is the regime-dependent interpretation of systemic importance. Financials and Real Estate remain recurrently important, but the most central stocks are not identical across crisis episodes. During domestic funding stress, systemic importance becomes more concentrated in the Real Estate and Financials nexus, reflecting refinancing pressure, collateral deterioration, margin-related selling, and weakening market liquidity. During external repricing shocks, the network rotates toward Materials, Energy, Consumer Staples, infrastructure-linked firms, and liquidity-sensitive financial intermediaries. This pattern implies that systemic importance is not a fixed attribute of firm size or market capitalization. A stock becomes systemically important when its network position and sectoral exposure match the dominant transmission channel of the crisis. This interpretation is consistent with the systemic-risk literature, which emphasizes that financial fragility depends on interconnectedness,

common exposure, and the contribution of individual institutions or assets to system-wide distress (Acharya et al., 2017; Adrian and Brunnermeier, 2016; Wang and Huang, 2021).

The comparison between Pearson correlation and Clayton lower-tail dependence should therefore be read as part of the same story. Pearson correlation remains useful for describing ordinary co-movement, but the results show that it does not fully identify the structure of co-crash risk. The low overlap between the Pearson and lower-tail PMFG networks indicates that normal co-movement and downside co-movement organize the market differently. More importantly, the crisis validation results show that SII_{tail} explains realized crisis-period losses more consistently than SII_{corr} , especially during the Bond Shock, when funding stress and collateral concerns were central to market deterioration. This finding confirms the economic relevance of the lower-tail network: it does not only produce a different network map, but also identifies stocks whose positions are associated with larger realized losses during stress periods (Joe, 2014; Nelsen, 2006; Wang and Huang, 2021).

Taken together, the findings answer the three research questions of the dissertation. First, HOSE-listed stocks exhibit meaningful lower-tail dependence, and this dependence intensifies under stress. Second, the stocks and sectors contributing most to market-wide downside risk are concentrated around the Financials and Real Estate core, but systemic importance rotates toward other cyclical and externally sensitive sectors depending on the nature of the shock. Third, lower-tail copula dependence captures crisis-period co-movement more effectively than Pearson correlation because it focuses on the part of the return distribution where diversification failure and systemic transmission are most relevant.

5.2 Implications

The findings have implications for market supervisors, policymakers, exchanges, asset managers, and investors. The main implication is that systemic-risk monitoring in

Vietnam should move beyond isolated indicators of volatility, liquidity, market capitalization, and average correlation. These indicators remain useful, but they do not fully capture whether downside losses are becoming synchronized across stocks and sectors. A market can experience moderate index losses while lower-tail dependence rises sharply, as shown by the U.S. reciprocal tariff shock. Conversely, a large market correction may involve weaker tail synchronization if losses are more heterogeneous across stocks. For this reason, lower-tail dependence and tail-based systemic importance can provide an additional layer of information for monitoring market fragility (McNeil et al., 2015; Wang et al., 2022).

For regulators and market supervisors, the results suggest that a lower-tail network dashboard could complement existing market surveillance tools. Such a dashboard would not replace price limits, liquidity monitoring, margin supervision, or disclosure requirements. Rather, it would help identify when co-crash risk is becoming concentrated in specific stocks or sectors. The most relevant indicators would include the level of market-wide lower-tail dependence, the concentration of strong lower-tail links, the sectoral composition of the tail network, and changes in SII_{tail} rankings across regimes. If the same sectors repeatedly appear as central nodes in the lower-tail network, supervisors may treat them as structural sources of downside-risk transmission. If new sectors suddenly become central, supervisors may interpret this as evidence of regime-specific risk rotation.

The Financials and Real Estate nexus deserves particular attention in macroprudential monitoring. The results indicate that this nexus is the most persistent core of crash transmission on HOSE. This does not imply that all banks, securities firms, or property developers are equally risky. Rather, it means that stress in this group can have broader market implications because the sectors are connected through credit conditions, collateral values, corporate bond confidence, margin lending, and investor de-risking. During periods of credit tightening, property-market stress, bond-market repayment concerns, or declining collateral values, regulators may therefore benefit from

monitoring joint movements across banks, securities companies, and real-estate firms, rather than assessing each group separately. This is especially relevant in Vietnam, where financial-sector connectedness and real-estate exposure have been shown to matter for systemic risk and market stress (Anh T. H. Nguyen and Thanh T. Le, 2025; Xuan T. T. Tram and Hoai T. T. Nguyen, 2021).

The results also have implications for policy communication and crisis management. When market stress is concentrated in the lower-tail network, uncertainty may spread through investor expectations before balance-sheet losses are fully observed. Clear communication on liquidity conditions, corporate bond restructuring, real-estate credit exposure, and market-stabilization measures may reduce the risk that investors interpret sector-specific stress as a signal of broader financial instability. This is particularly important for sectors that serve as market barometers, such as banks, securities firms, and large real-estate developers. In this sense, the lower-tail network is not only a statistical tool. It can help policymakers understand which market segments require clearer information, closer supervision, or coordinated monitoring during stress periods.

For the stock exchange and market infrastructure institutions, the findings suggest that systemic-risk analysis should consider the cross-sector structure of downside co-movement. Market surveillance often focuses on abnormal price movements, trading volume, liquidity deterioration, and compliance issues at the individual-stock level. The evidence in this dissertation shows that an additional network perspective is useful because systemic stress may arise from groups of stocks that become jointly vulnerable. Monitoring whether strong lower-tail links are becoming concentrated within Financials, Real Estate, Materials, Energy, or infrastructure-related stocks can help identify periods when market stress is no longer isolated. This perspective may also support scenario analysis, stress testing, and sector-based risk reporting.

For investors and portfolio managers, the key implication is that diversification should be evaluated under downside states, not only under average market conditions. A

portfolio that appears diversified using Pearson correlation may still be exposed to common lower-tail losses if it holds stocks connected through credit, collateral, liquidity, commodity, or investment-cycle channels. This is particularly relevant for portfolios concentrated in Financials, Real Estate, Materials, Energy, or other high-beta cyclical stocks. Investors should therefore combine conventional correlation analysis with downside-dependence measures, especially when designing stress tests, setting risk limits, or assessing portfolio drawdown exposure. The purpose is not to avoid all systemically important stocks, but to recognize when several holdings may become vulnerable to the same adverse state (Anh T. Le and Binh T. T. Dao, 2021; Low, 2015; Markowitz, 1952).

The regime-dependent results also imply that risk management should be dynamic. Static measures such as market capitalization, beta, or normal-period correlation may fail to identify stocks that become central under a specific shock. During domestic funding stress, investors should pay closer attention to property developers, banks, securities firms, margin-sensitive stocks, and firms exposed to refinancing pressure. During external repricing shocks, attention should shift toward Materials, Energy, export-linked Consumer Staples, infrastructure-related firms, and liquidity-sensitive financial intermediaries. This does not mean that each sector will always decline in the same way. Rather, the relevant point is that the channel of systemic risk changes with the origin of the shock. Portfolio monitoring should therefore combine sector exposure, funding conditions, liquidity indicators, and lower-tail connectedness.

Finally, the findings have a broader implication for financial-stability research in emerging markets. Vietnam's stock market illustrates how systemic risk can arise from the interaction between financial development, sectoral concentration, market liquidity, and investor behavior. Lower-tail dependence is useful because it connects these economic features to observable market prices. The framework developed in this dissertation can therefore support future research and practical monitoring by linking

stock-level dependence, sectoral transmission channels, and crisis-period losses within a single empirical structure.

5.3 Limitations and Future Research

Several limitations should be acknowledged. First, the analysis is limited to HOSE-listed stocks. HOSE represents the main equity market in Vietnam and includes many large and liquid firms, but it does not cover the full listed universe. Future research could extend the sample to HNX and UPCoM to examine whether lower-tail dependence differs across exchanges with different liquidity profiles, firm characteristics, and investor participation.

Second, the study uses pairwise Clayton copulas to focus on lower-tail dependence. This choice is appropriate for the research question because the dissertation is concerned with joint downside movements. However, pairwise copulas do not fully capture higher-dimensional dependence among multiple stocks. Future studies could use vine copulas, dynamic copulas, or other multivariate dependence models to examine whether systemic risk is transmitted mainly through bilateral links, sectoral clusters, or broader multi-stock structures.

Third, the regime-based design allows the study to compare normal periods and selected stress episodes, but it does not fully describe the continuous evolution of systemic risk. A time-varying network approach could trace how lower-tail dependence builds up before crises, peaks during stress, and normalizes afterward. This would be useful for identifying early-warning indicators of downside fragility.

Fourth, the interpretation of transmission channels is based primarily on market prices and sectoral economic reasoning. The analysis does not directly include firm-level balance-sheet variables, ownership linkages, corporate bond exposure, collateral data, or margin-lending information. Future research could combine market-based lower-tail networks with firm-level financial data to identify more precisely why certain stocks become systemically important during specific regimes.

These limitations define the scope of the study, but they do not weaken its main conclusion. The evidence shows that lower-tail dependence provides useful information about crash transmission in Vietnam's stock market. Future research can extend this framework by using broader samples, richer firm-level data, and more dynamic modelling approaches.

5.4 Conclusion

This dissertation examines systemic risk and lower-tail dependence in Vietnam's stock market using a copula-based network approach. The main conclusion is that systemic risk on HOSE is not only a matter of which stocks fall the most, but also a matter of which stocks fall together and occupy central positions in the downside-dependence network. The results show that lower-tail dependence is present among HOSE-listed stocks, becomes stronger in stress regimes, and is concentrated around economically meaningful sectoral channels.

The Financials and Real Estate nexus forms the most persistent source of downside-risk transmission. This reflects the interaction between credit conditions, collateral values, refinancing pressure, margin lending, market liquidity, and investor confidence. At the same time, Materials, Energy, Utilities, infrastructure-related firms, and selected consumer or export-linked stocks become more systemically relevant when shocks are transmitted through commodity prices, construction demand, external repricing, or trade-related uncertainty. Systemic importance is therefore both persistent and regime-dependent. Some sectors form a stable core, while others become central when the nature of the shock changes.

The comparison with Pearson correlation reinforces this conclusion. Correlation-based networks describe average co-movement, but they do not fully capture the structure of crash-period dependence. The lower-tail network identifies different links and provides stronger explanatory power for realized crisis-period losses. Overall, the dissertation shows that systemic risk in Vietnam's stock market is best understood as a

regime-dependent lower-tail network shaped by sectoral linkages, liquidity conditions, credit and collateral channels, and crisis-specific repricing mechanisms. This perspective provides a clearer finance narrative for why diversification can fail during crises and why certain sectors become systemically important in an emerging equity market.

References

- Acemoglu, D., Ozdaglar, A., Tahbaz-Salehi, A., 2015. Systemic Risk and Stability in Financial Networks. *Am. Econ. Rev.* 105, 564–608. <https://doi.org/10.1257/aer.20130456>
- Acharya, V.V., Pedersen, L.H., Philippon, T., Richardson, M., 2017. Measuring systemic risk. *Rev. Financ. Stud.* 30, 297–365. <https://doi.org/10.1093/rfs/hhw008>
- Adrian, T., Brunnermeier, M.K., 2016. CoVaR. *Am. Econ. Rev.* 106, 1705–1741. <https://doi.org/10.1257/aer.20120555>
- Allen, F., Gale, D., 2000. Financial Contagion. *J. Polit. Econ.* 108, 1–33. <https://doi.org/10.1086/262109>
- Anh P. Nguyen, P.A., Nguyen, N.P., Le, H.M.D., 2023. Systemic risk and financial system network using financial risk meter: the case of Vietnam. *Appl. Econ.* 56. <https://doi.org/10.1080/00036846.2023.2174936>
- Anh T. H. Nguyen, A.T.H., Thanh T. Le, T.T., 2025. In bank runs and market stress, it matters how networks impact: Exploring the financial connectedness in Vietnam. *Finance Res. Lett.* 72, 106489. <https://doi.org/10.1016/j.frl.2024.106489>
- Anh T. Le, T.A., Binh T. T. Dao, T.T.B., 2021. Portfolio optimization under mean-CVaR simulation with copulas on the Vietnamese stock exchange. *Invest. Manag. Financ. Innov.* 18, 273–286. [https://doi.org/10.21511/imfi.18\(2\).2021.22](https://doi.org/10.21511/imfi.18(2).2021.22)
- Anh T. T. Tran, T.T.A., 2023. Risk spillover from the US stock market into Vietnam: empirical evidence by VAR-DCC-GARCH and quantile regression. *J. Econ. Dev.* 15–27. <https://doi.org/10.33301/JED.VI.1095>
- Bekaert, G., Ehrmann, M., Fratzscher, M., Mehl, A., 2014. The global crisis and equity market contagion. *J. Finance* 69, 2597–2649. <https://doi.org/10.1111/jofi.12203>
- Brownlees, C., Engle, R.F., 2017. SRISK: A Conditional Capital Shortfall Measure of Systemic Risk. *Rev. Financ. Stud.* 30, 48–79. <https://doi.org/10.1093/rfs/hhw060>
- Chien V. Nguyen, V.C., Thuy T. Nguyen, T.T., 2022. Dependence between Chinese stock market and Vietnamese stock market during the Covid-19 pandemic. *Heliyon* 8, e11090. <https://doi.org/10.1016/j.heliyon.2022.e11090>
- Cuong Nguyen, C., Bhatti, M.I., Henry, D., 2017. Are Vietnam and Chinese Stock Markets Out of the US Contagion Effect in Extreme Events? *Phys. Stat. Mech. Its Appl.* 480, 10–23. <https://doi.org/10.1016/j.physa.2017.02.045>
- Diebold, F.X., Yilmaz, K., 2012. Better to give than to receive: predictive directional measurement of volatility spillovers. *Int. J. Forecast.* 28, 57–66. <https://doi.org/10.1016/j.ijforecast.2011.02.006>
- Duong, D., Huynh, T.L.D., 2020. Tail dependence in emerging ASEAN-6 equity markets: empirical evidence from quantitative approaches. *Financ. Innov.* 6. <https://doi.org/10.1186/s40854-019-0168-7>
- Duy Duong, D., Duc T. L. Huynh, T.L.D., 2020. Tail Dependence in Emerging ASEAN-6 Equity Markets: Empirical Evidence from Quantitative Approaches. *Financ. Innov.* 6, 1–29. <https://doi.org/10.1186/s40854-019-0168-7>

- Joe, H., 2014. *Dependence Modeling with Copulas*. Chapman and Hall/CRC, Boca Raton. <https://doi.org/10.1201/b17116>
- Li, X., Rose, L.C., 2009. The tail risk of emerging stock markets. *SSRN Electron. J.* <https://doi.org/10.2139/ssrn.1413463>
- Low, R., 2015. Vine copulas: modelling systemic risk and enhancing higher-moment portfolio optimisation. *Account. Finance*. <https://doi.org/10.1111/acfi.12274>
- Markowitz, H., 1952. Portfolio Selection. *J. Finance* 7, 77–91. <https://doi.org/10.2307/2975974>
- McNeil, A.J., Frey, R., Embrechts, P., 2015. *Quantitative Risk Management: Concepts, Techniques and Tools*, Revised. ed. Princeton University Press, Princeton.
- Mensi, W., Ziadat, S.A., Vo, X.V., Kang, S.H., 2024. Extreme quantile connectedness and spillovers between oil and Vietnamese stock markets: a sectoral analysis. *Int. J. Emerg. Mark.* 19, 1586–1625. <https://doi.org/10.1108/IJOEM-03-2022-0513>
- Nelsen, R.B., 2006. *An Introduction to Copulas*, 2nd ed. Springer, New York. <https://doi.org/10.1007/0-387-28678-0>
- Nhat T. H. Dang, T.H.-N., Nguyen, N.T., Vo, D.H., 2023. Sectoral volatility spillovers and their determinants in Vietnam. *Econ. Change Restruct.* 56, 681–700. <https://doi.org/10.1007/s10644-022-09446-9>
- Patton, A.J., 2012. A review of copula models for economic time series. *J. Multivar. Anal.* 110, 4–18. <https://doi.org/10.1016/j.jmva.2012.02.021>
- Sklar, A., 1959. Fonctions de répartition à n dimensions et leurs marges. *Publ. Inst. Stat. Univ. Paris* 8, 229–231.
- Thuy T. Nguyen, T.T., Chien V. Nguyen, V.C., Trong T. Nguyen, T.T., 2020. Oil price shocks against stock return of oil- and gas-related firms in the economic depression: new evidence from a copula approach. *Cogent Econ. Finance* 8, 1–20. <https://doi.org/10.1080/23322039.2020.1799908>
- Tumminello, M., Aste, T., Di Matteo, T., Mantegna, R.N., 2005. A Tool for Filtering Information in Complex Systems. *Proc. Natl. Acad. Sci.* 102, 10421–10426. <https://doi.org/10.1073/pnas.0500298102>
- Van T. T. Vu, T.T.V., Kham D. Tran, D.K., 2019. Systemic risk in Vietnam stock market. *Asian Econ. Financ. Rev.* 9, 339–352. <https://doi.org/10.18488/journal.aefr.2019.93.339.352>
- Wang, D., Huang, W.-Q., 2021. Centrality-based measures of financial institutions' systemic importance: a tail dependence network view. *Phys. Stat. Mech. Its Appl.* 562, 125345. <https://doi.org/10.1016/j.physa.2020.125345>
- Wang, Z., Gao, X., Huang, S., Sun, Q., Chen, Z., Tang, R., Di, Z., 2022. Measuring systemic risk contribution of global stock markets: A dynamic tail risk network approach. *Int. Rev. Financ. Anal.* 84, 102361. <https://doi.org/10.1016/j.irfa.2022.102361>
- Wen, F., Yang, X., Zhou, W., 2018. Tail dependence networks of global stock markets. *Int. J. Finance Econ.* 24, 558–567. <https://doi.org/10.1002/ijfe.1679>

- Xuan T. T. Tram, T.T.X., Hoai T. T. Nguyen, N.T.T., 2021. Effect of the macroeconomic variables on systemic risk: evidence from the Vietnamese economy. *Econ. Bus. Lett.* 10, 217–228. <https://doi.org/10.17811/ebl.10.3.2021.217-228>
- Zhou, C., Qin, X., 2021. Time-varying asymmetric tail dependence of international equities markets. *Pac.-Basin Finance J.* 68, 101589. <https://doi.org/10.1016/j.pacfin.2021.101589>